
TC 1703

Operating and Maintenance Manual

Customer
Power Station
Pelton Turbine

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TC 1703
Operating and Maintenance
Manual
Pelton Turbine

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 TC 1703 and TC 1703 XL

1. Introduction

These Operating and Maintenance Instructions are intended for the technical personnel of the operator and cover the description of the:

- Functionality of the controller program,
- Operating via the touch panel,
- Parameters which can be changed from the touch panel, and the
- Error messages which are shown on the touch panel.

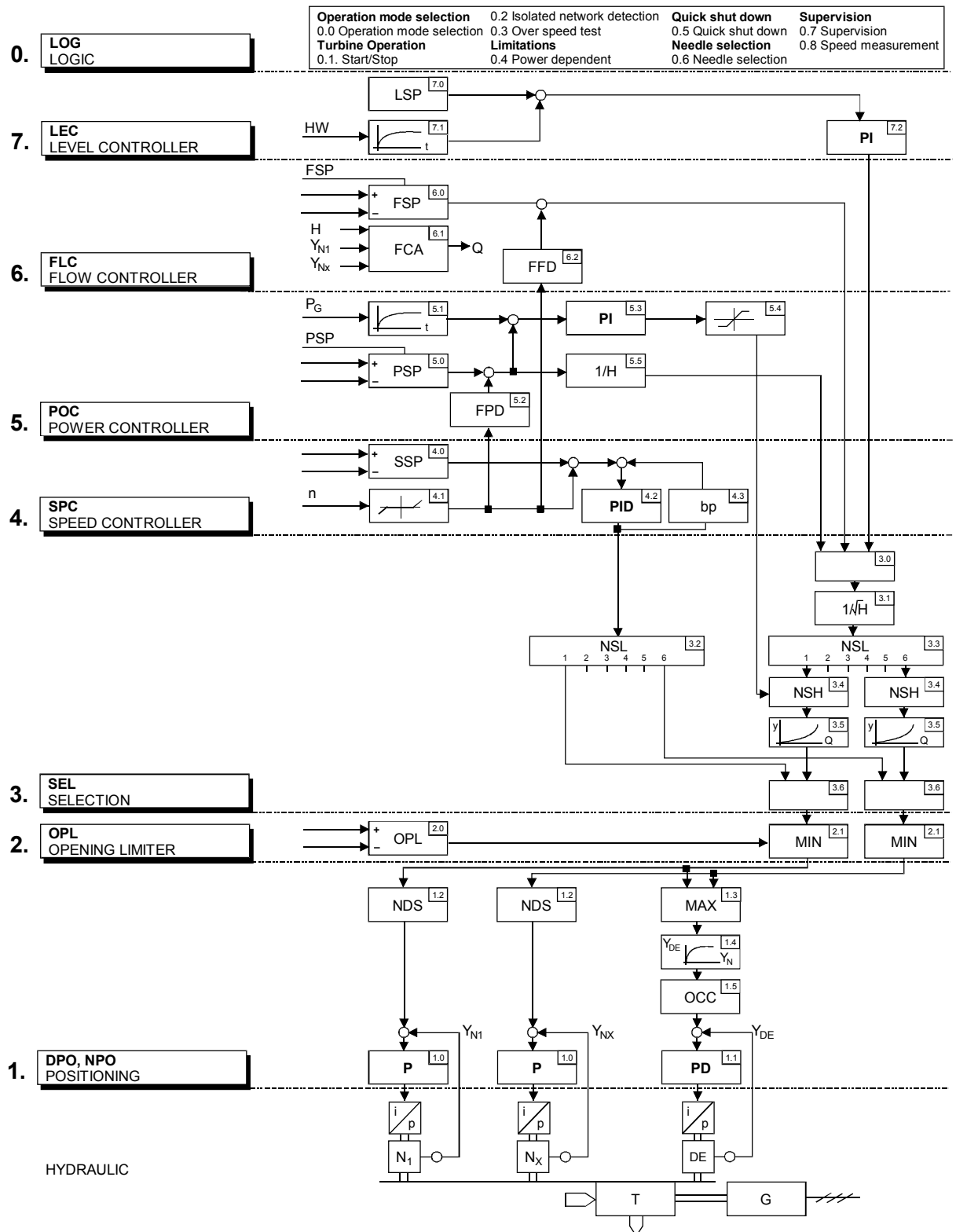
2. Abbreviations

DE	D eflector
DPO	D eflector P ositioning
FCA	F low C alculation
FFD	F requency F low D roop
FLC	F low C ontroller
FPD	F requency P ower D roop
FSP	F low S et P oint
G	G enerator
LEC	L evel C ontroller
LOG	L OGic
LSP	L evel S et P oint
N	N eedle
NDS	N eedle D i S charge
NPO	N eedle P ositioning
NSH	N eedle S Hifting
NSL	N eedle S e L ection
OCC	O ff C am C ontrol
OPL	O Pening L imiter
POC	P ower C ontroller
PSP	P ower S et P oint
SEL	S E L ection Controller Setpoint
SPC	S Peed C ontroller
SSP	S peed S et P oint
T	T urbine
TC	T urbine C ontroller

Definitions

KP	proportional gain
TN	integral-action time
TD	derivative-action time
bp	Permanent droop
G(s)	transfer function
H	head
HW	level headwater
n	speed
P _G	active power
Q	flow
Y _{DE}	deflector position
Y _{Nx}	needle position

3. Block Diagram - Pelton Turbine



Functions

0. Logic

Operating mode selection

0.0 Operating mode selection

Turbine operation

0.1 Start/stop

0.2 Isolated Network Detection

0.3 Overspeed test

Limitations

0.4 Power dependent

Quick shutdown

0.5 Quick shutdown

Needle selection

0.6 Needle selection

Monitorings

0.7 Monitoring

0.8 Speed measuring

1. Positioning

1.0 Positioning - Needles

1.1 Positioning - Deflector

1.2 Needle discharge

1.3 Maximum needle opening setpoint value

1.4 CAM (needle/deflector relationship)

1.5 Deflector off-cam control
(interconnected operation)

2. Opening limiting

2.0 Setpoint adjuster

2.1 Limitation of Controller Output

3. Selection of the Opening Setpoint Value

3.0 Selection (Interconnected operation)

3.1 Head influence

3.2 Needle selection - Startup/isolated
network operation

3.3 Needle selection - Interconnected
operation

3.4 Needle shifting

3.5 Determination of the needle position
from the flow

3.6 Selection - Startup/isolated network
operation - interconnected operation

4. Speed Controller

4.0 Setpoint adjuster

4.1 Frequency dead band

4.2 Speed controller (parallel PID structure)

4.3 Permanent droop

5. Power Controller

5.0 Setpoint adjuster

5.1 Input filter

5.2 Frequency/power droop

5.3 Power controller (parallel PI structure)

5.4 Output limiting

5.5 Conversion of power setpoint value to
flow setpoint value

6. Flow Controller

6.0 Setpoint adjuster

6.1 Flow calculation

6.2 Frequency/flow droop

7. Level Controller

7.0 Setpoint adjuster

7.1 Input filter

7.2 Level controller (parallel PI structure)

4. Description of the Control Functions

4.1 General

The controller application for the multi-nozzle Pelton turbine has a parallel structure consisting of speed, flow, level and power controlling. In each case, the controller signal pertaining to the selected controller operating mode will be selected and fed to the downstream positioning circuit. During this period, the other controllers are adjusted in a way that bumpless changeover can take place any time. The speed controller is automatically activated during the synchronization or in isolated operation.

The structure of the controller application is arranged in such a way that the main functions (speed, flow, level and power controlling) are available as separate parts of the program which can be adapted to the plant at any time.

The following pages describe levels 0 to 7 of the block diagram shown in Chapter 3.

4.2 Level 0 - Logic

The various operating modes [0.0] (speed, flow, level and power control) are, according to the selection, checked for plausibility and allocated. In the event of a fault, a switchover is made to another possible operating mode (flow control) or the machine is shut down.

The start/stop logic [0.1] enables automatic starting (at nominal speed) and shutting down of the machine. As standard, the unit is started up with two nozzles. With the command «TC Start» - as long there is a minimum value for the opening limiter and no «TC Stop» command – the opening limiter is set to the start opening. After synchronization of the machine the opening limiter is set to maximum.

With the «TC Stop» command, the opening limiter is set equal to the maximum needle position and then returned to the minimum value with the set runtime.

In the event of load rejection, the setpoint adjusters are automatically reset so that synchronization can be done at any time (Speed setpoint value = 100%, Opening limiter = 50%).

The internal isolated network detection [0.2] monitors frequency deviations from the nominal value (50/60 Hz = 100%). When an upper or lower limit value (outside the speed band) responds, the message «Internal isolated network detection» is output as long as the «External isolated network detection» output is not present. Isolated network operation remains selected as long as "Internal isolated network detection" is not reset or the "External isolated network detection" input is not present. Generally, in isolated mode a changeover to speed control is carried out. The isolated network parameters (KP, TN, bp) are selected and, as standard, all nozzles are switched in.

When idling, the machine can be run to overspeed with two nozzles via an operating panel in order to test the electrical and mechanical overspeed protection trippings [0.3]. The maximum actual speed value is stored and is shown on the operating panel.

In order to prevent overloading of the generator, the nozzle opening is limited depending on the power [0.4] with the help of a PI controller. In the opening limiter, the currently permitted needle position is ascertained via minimum selection [2.1] and the active controller output is limited with this value.

On «Quick shutdown» [0.5] the opening limiter is set to minimum in one step.

The needle selection [0.6] puts the optimum number of nozzles into operation according to the operating point. As not only the flow controller but also the level and power controllers generate a flow setpoint value, for these operating modes the flow setpoint value applies as the switchover criterion of the nozzles. The nozzle with the highest priority is input externally. The remaining nozzles are switched-in in such a way that the bearing load is as symmetrical as possible.

The computer is itself monitored, in that the supply voltages and the program run are continuously checked (Watchdog function). Besides this, the status of the binary and analog input/output channels, the positioning circuits, the activated tasks and their cycle times are monitored [0.7].

For speed measuring, two analog speed inputs are used which are continuously monitored [0.8]. The second input evaluates the frequency of the terminal voltage of the generator winding. The signal of this measuring is, as standard, used inside the controller as long as it is not faulty. If it should be faulty, a switchover is automatically made to the other measuring. On failure of both measurements, quick shutdown is triggered by the output of "Failure of primary measuring signals».

4.3 Level 1 - Positioning

The positioning of the deflector servomotor is done via a PD controller [1.1] and the positioning of the needle servomotors is done in each case via a P controller [1.0]. The outputs of the positioning circuits activate the relevant servovalve. The positioning circuits are continuously monitored and a fault is reported. In addition, on malfunction of the deflector positioning circuit with the message «Deflector positioning faulty», emergency shutdown is triggered. The servomotors can be positioned manually for test purposes with the help of the operating panel.

When the machine is shut down, the spiral case may be emptied via the needle discharge. The servomotor opening of the needles and the time period may be freely adjusted.

The switching in of the needles is done by the needle selection automation [0.6, 3.2, 3.3, 3.4].

The deflector setpoint value is determined via the cam [1.4] from the maximum needle setpoint value [1.3]. When starting up and for isolated operation, it is always run in CAM.

In interconnected operation, the deflector can be taken out of cam [1.5]. In this case, the deflector is positioned to the maximum opening (= 100%). Thus, with fast load changes, it never intervenes in the jet.

4.4 Level 2 - Opening Limitation

The opening limiter [2.0] can, as standard, be set between –5 and 105%. The current value of the opening limiter limits the needle setpoint value via minimum selection [2.1].

4.5 Level 3 - Opening Setpoint Selection

In interconnected operation, according to the chosen operating mode, the calculated flow setpoint signal is selected by the flow, level or power controllers in the selection logic [3.0], converted with the current head [3.1] and switched through to the needle selection [3.3]. The needle shifting [3.4] guarantees the smooth switching in and out of needles. The position setpoint values of the needles are determined in accordance with the flow characteristic curve of the nozzles [3.5].

In the selection logic [3.6] a choice is made between the opening setpoint value of the speed controller (startup or isolated network operation) and the controller signal which is determined in the selection logic [3.0].

The flow setpoint of the inactive controller modes is corrected to the current actual flow value so that, at any time, a bumpless changeover can be made. For the flow, level and power controllers, the setpoint value is adapted to the actual value so that, as a consequence, the switchover does not result in any movement of the needles.

On the contrary, during the transition to isolated operation, there is generally a controlling deviation at the input of the speed controller which the controller can compensate only by readjusting of the needles and intervention of the deflector so that the desired setpoint speed is achieved.

4.6 Level 4 - Speed Controller

The speed setpoint [4.0] can, as standard, be set between 90 and 110%. On the start command and after synchronization, the setpoint is automatically reset to 100%.

In order to enable optimal synchronization for all operating points, on startup the influence of the permanent droop is automatically compensated in the speed controller.

In interconnected operation, the frequency dead band [4.1] is effective. The dead band limit can be freely set.

Using the frequency setpoint/actual value deviation, the speed controller [4.2] calculates the setpoint for the needle servomotor position. It is designed as a PID controller in a parallel structure. The permanent droop bp [4.3] determines the influence of the frequency changes on the controller output (isolated operation). The transfer function of the speed controller on neglect of the time-delay element in the differential element is:

$$G(s) = \frac{KP \left(1 + \frac{1}{TN \cdot s} + TD \cdot s \right)}{1 + KP \left(1 + \frac{1}{TN \cdot s} + TD \cdot s \right) \cdot bp}$$

or, with permanent droop $bp \approx 0$

$$G(s) = KP \left(1 + \frac{1}{TN \cdot s} + TD \cdot s \right).$$

4.7 Level 5 - Power Controller

The power setpoint [5.0] can, as standard, be set between –5 and 105%. The power controller operating mode can only be selected in interconnected operation.

The power setpoint can be altered either via higher/lower commands or an external power setpoint is input.

The active power signal is led via an input filter [5.1]. The filter time constant can be freely set for this. On signal failure, an alarm is output and a switchover is automatically made to flow control.

The power controller is executed as a correction controller with feedforward control. With the power setpoint as the feedforward signal, the needles are selected via the needle selection [3.3] and the setpoint for the needle servomotor positions are determined. The power setpoint value is converted into a flow setpoint value taking account of the current head [5.5]. The power correction controller is designed as a PI controller in a parallel structure [5.3]. The transfer function is:

$$G(s) = KP \cdot \left(1 + \frac{1}{TN \cdot s} \right).$$

The output of the power correction controller is limited by the output limiter [5.4].

On frequency deviations from the nominal value, the amount for the frequency back-up is determined via the frequency-power droop [5.2] and added to the power setpoint value.

4.8 Level 6 - Flow Controller

The flow setpoint [6.0] can, as standard, be set between –5 and 105%. The flow controller operating mode can only be selected in interconnected operation.

The flow setpoint can be altered either via higher/lower commands or an external flow setpoint is input.

Using the head and the nozzles which are in operation, the current actual flow is calculated [6.1] with the help of the current needle positions. This value is used for tracking of the flow and level controllers.

On frequency deviations from the nominal value, the amount for the frequency back-up is determined via the frequency-flow droop [6.2] and added to the flow setpoint value.

4.9 Level 7 - Level Controller

The desired level setpoint [7.0] is pre-input as a parameter value and can be changed via the operating panel. The level control operating mode can only be selected in interconnected operation. So that, on activation of the level controller only minimal flow and level changes occur, the setpoint for the controller is run via a gradient limiter to the desired level setpoint based on the current actual value.

The level signal is led via an input filter [7.1]. The filter time constant can be freely set. On signal failure, the last valid value is stored, a warning is output, and a switchover to flow control takes place automatically.

The level controller [7.2] calculates the flow setpoint using the level setpoint / actual value difference. It is designed as a PI controller in a parallel structure. The transfer function of the level controller is:

$$G(s) = KP \cdot \left(1 + \frac{1}{TN \cdot s} \right)$$

When using an external level controller (joint controller), the flow setpoint is read-in either as an analog input signal or via a communications interface.

5. Functionality of the 15" Touch Panel

5.1 Fundamental Structure

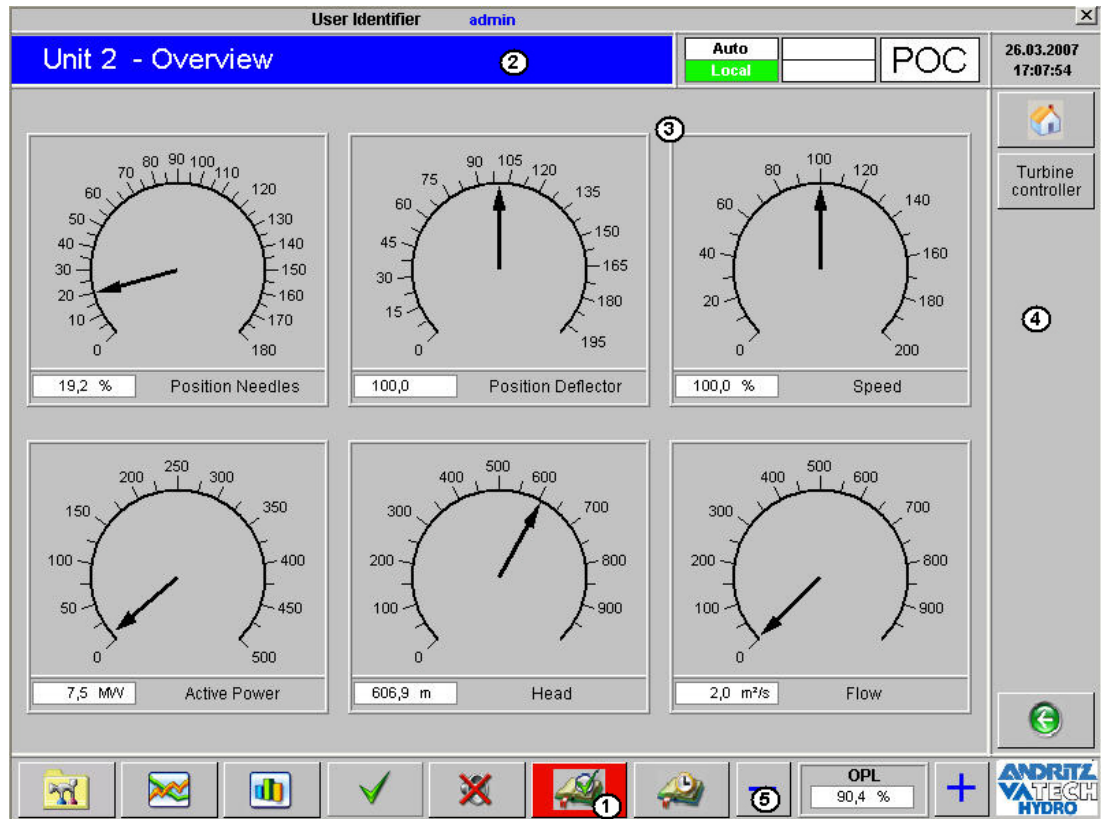


Fig. 1: Principle Design 15"-Panel Pelton turbine

The fundamental structure of the 15" touch panel for the Pelton turbine is structured into the following five parts:

- **Alarm and status window (1)**
In the center of this window, the current operating mode of the turbine controller is displayed. On the left side, the free memory space of the memory card is indicated, while on the right side the free memory space of the internal panel memory is shown. In the event of an alarm or a warning, this status window will be hidden by the alarm window. The latter shows the last occurring alarm or the last occurring warning against a red background, until it is acknowledged by the user:

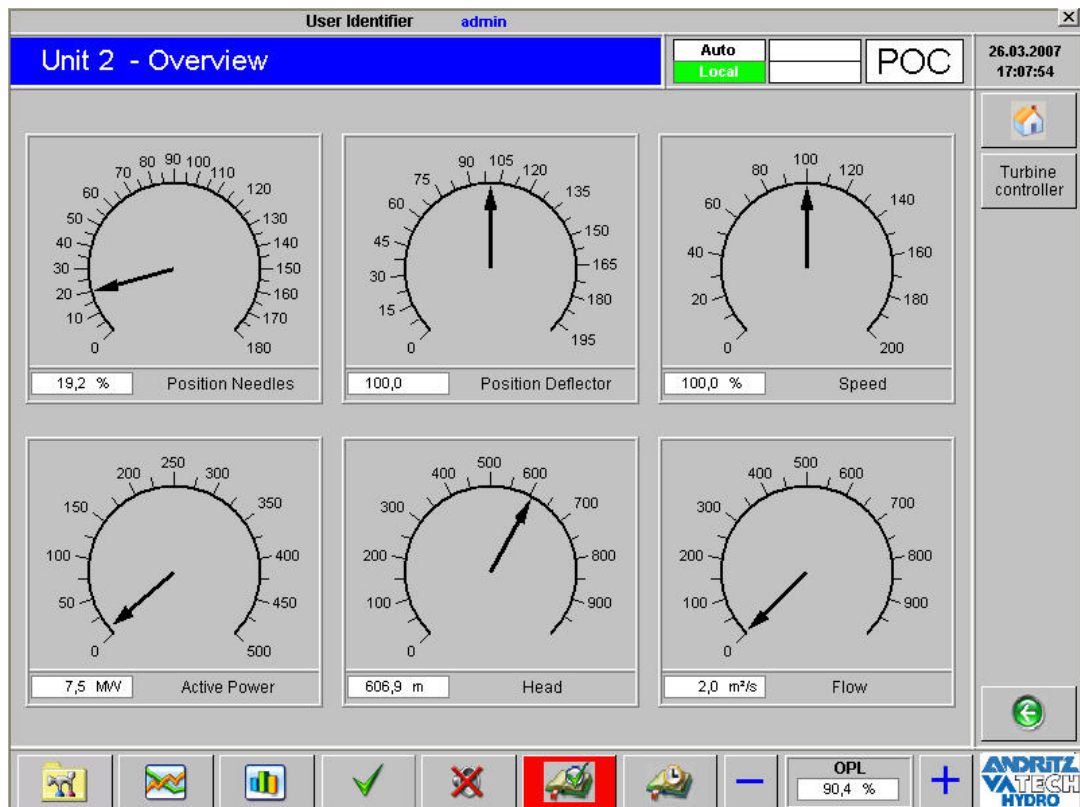


Fig. 2: Display – Alarm line

The status window first becomes visible again when all alarms or warnings have been acknowledged by the user in the Alarm Signal List.

- **Header (2)**
Here, besides the current display title, the data and time of the system is also displayed. At the left side of time and date some additional items will be displayed:
 - automatical or manual operation
 - local or remote operation
 - island detection (internal or external)
 - turbine in limited state
 - current active controller (POC- Power Controller, LEC – Level Controller, OPC- Opening Controller, SPC – Speedcontroller, FLC-Flow Controller)
- **Main window (3)**
Here the respective operating- and monitoring windows are superimposed.
- **Navigation menu (4)**
The display navigation is carried out using the buttons of this menu. There is a navigation menu for each display, that displays all pictures that can be called from this display.
This Touch-Panel program contains five display levels, of which one, the lowest level in the hierarchy, can only be reached via the supplementary menu.
Some navigation menus are structured in three pages in order to accommodate the large number of detailed displays.

— Main menu buttons



- main overview display



- Back to last open display



- next page



- previous page

- Supplementary menu (5)

The additional images can be opened using the buttons of this menu. This supplementary menu is available in this form in all displays, with the exception of the supplementary displays themselves.

— Supplementary menu buttons



- System control



- Trend diagram



- Overview of measured values



- quit isolated network operation



- quit horn



- Alarm Signal List



- Chronologic Event List

(+) and

(-) changes to the opening limiter at any time

On the detailed controller displays, the feedback sensor calibration and the commissioning display there are numerous parameters, that one can change from the Touch-Panel. For this one clicks on the corresponding setpoint, which opens the following dialogue:

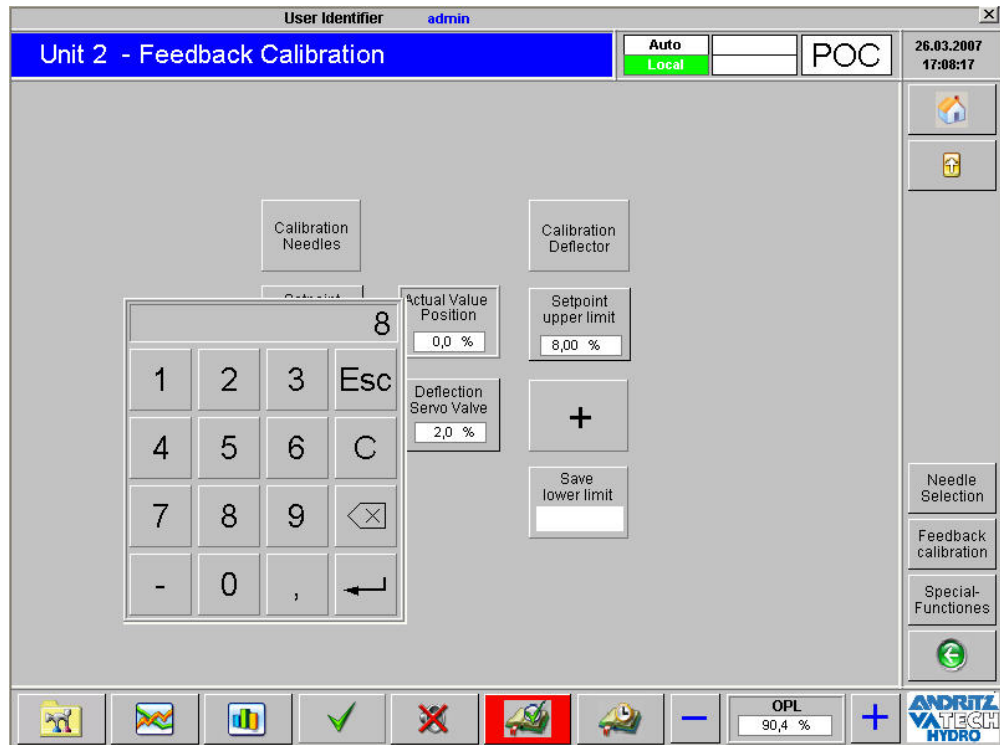


Fig. 3: Setpoint input

The desired setpoint value can be entered via the virtual keyboard on the touch panel. Once the entry has been confirmed by actuating the ↵ button, the value will be applied immediately. The entry can be corrected any time by actuating the ← button, or cancelled by actuating the ESC button.

5.2 Display – Overview

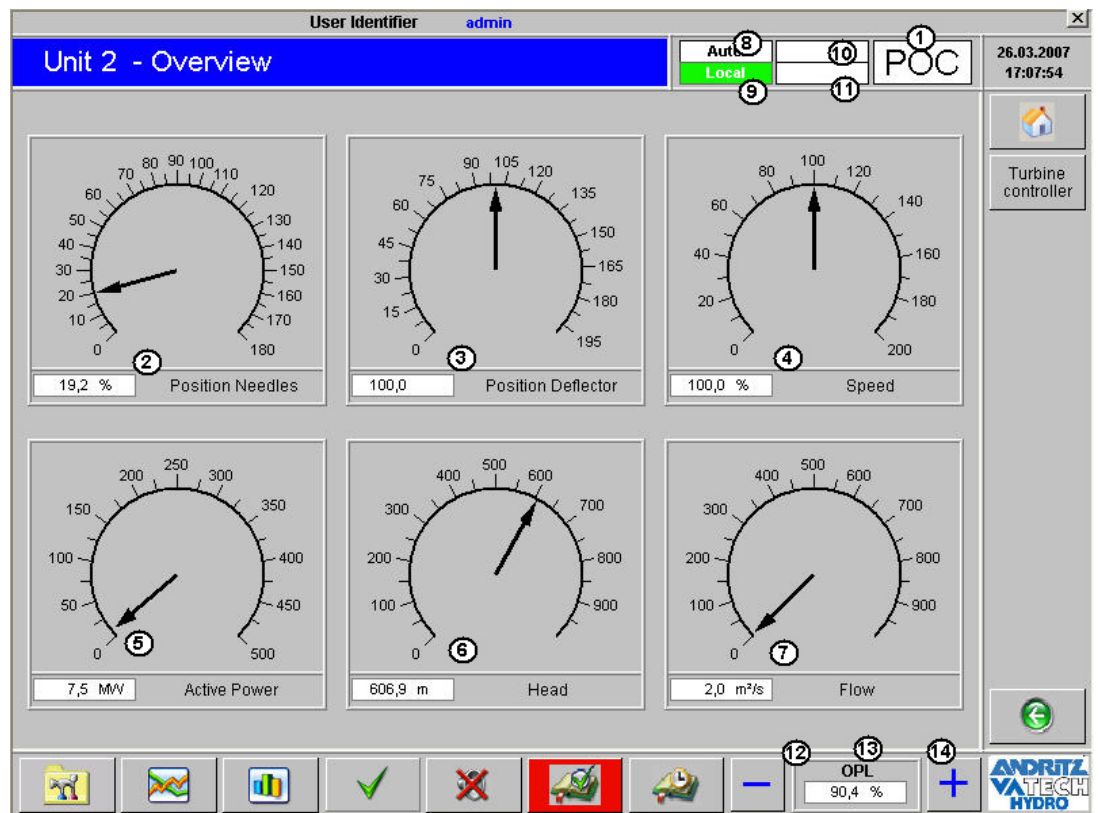


Fig. 4: Display – Overview

By default, this main overview display contains the most important machine data and -information. The main window itself contains only display elements and no operating elements. This overview is activated automatically after the start of the SAT250-Runtime. One can reach this display from any other display by pressing the -button in the navigation menu. Since with this display it has to do with the highest display level, both the -button as well as the -button are deactivated and are therefore displayed gray. From this display, using the navigation menu one reaches the controller overview display described in the following.

In the active state, the machine information (8 to 10) appear on a green background. The analog values (2 to 7) contained in this display can be found again in the trend diagram () and in the overview of measured values (), that one reaches by means of the supplementary menu.

Optionally, this main window can have a plant-specific representation or drawing superimposed on it.

Variables shown:

No.	Variable	Description
1	P.NW.DO.SPC_ON_INT	Display of working regulator
2	P.NW.AO.N_MAX	Position needles [%]
3	P.NW.AO.DE_INT	position deflector [%]
4	P.NW.AO.n_ACTIVE	speed [%]
5	P.NW.AO.P_INT	active power [MW]
6	P.NW.AO.H	head [m]
7	P.NW.AO.Q	flow [m³/s]
8	P.NW.DO.MAN_MODE	Hand / automatic mode
9	P.NW.DO.LOCAL	Local / remote
10	P.NW.DO.ISNET	Island operation
11	P.NW.DO.TU_LIMITED	turbine limited
12	P.NW.DI.OPL-	opening limit lower
13	P.NW.AO.OPL	opening limit [%]
14	P.NW.DI.OPL+	opening limit higher

Tab 1: Variables – Display Overview

5.3 Display – Controller Overview

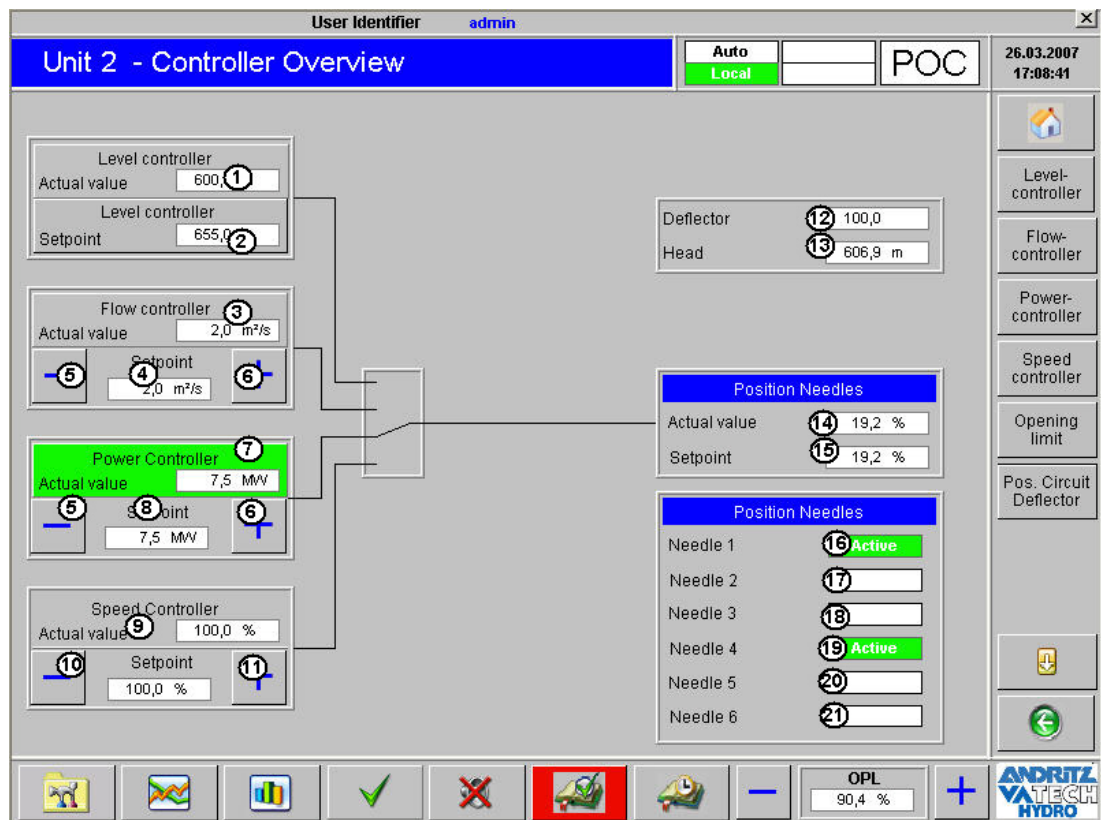


Fig. 5: Display – Controller Overview

This controller overview display combines the most important values of the machine and the associated controller in one single Figure. In this main window, the controller operating mode can be switched over to another one. To do so, actuate the respective controller buttons (1, 3 or 7). The speed controller can not be activated explicitly, as it is activated automatically depending on the operating state of the machine. The pertinent controller setpoints can be increased or lowered, as necessary, by actuating the buttons (+) and (–) (5 and 6 or 10 and 11). The setpoints are moved in the desired direction as long as the respective button is kept actuated. The pertinent readjustment gradients can be adapted on the associated detailed controller displays. In addition, this display indicates the most important machine data. The state of the needles is also shown on this display.

From this display, you can open all detailed controller displays and the feedback sensor calibration as well as the commissioning display via the navigation menu. The selection buttons for the two latter displays are on the third page of the menu.

—  - Next page

Variables shown:

No.	Variable	Description
1	P.NW.DI.LEC_ON P.NW.AO.HHW1_INT	level controller on headwater level [m]
2	P.NW.AO.LSP	level setpoint [m]
3	P.NW.DI.FLC_ON P.NW.AO.Q	flow controller on flow [m ³ /s]
4	P.NW.AO.FSP	flow setpoint [m ³ /s]
5	P.NW.DI.PSP/FSP-	power/flow setpoint lower
6	P.NW.DI.PSP/FSP+	power/flow setpoint higher
7	P.NW.DI.POC_ON P.NW.AO.P_INT	power controller on active power [MW]
8	P.NW.AO.PSP	power setpoint [%]
9	P.NW.AO.n_ACTIVE	speed [%]
10	P.NW.DI.SSP-	speed setpoint lower
11	P.NW.DI.SSP+	speed setpoint higher
12	P.NW.AO.DE_INT	Position deflector [%]
13	P.NW.AO.H	head [m]
14	P.NW.AO.N_MAX	maximum needle position [%]
15	P.NW.AO.SP_OPL_MAX	maximum needle setpoint [%]
16-21	P.NW.DO.Nx_ON_INT	Needle x active

Tab 2: Variables – Display Controller Overview

5.4 Display – Level Controller

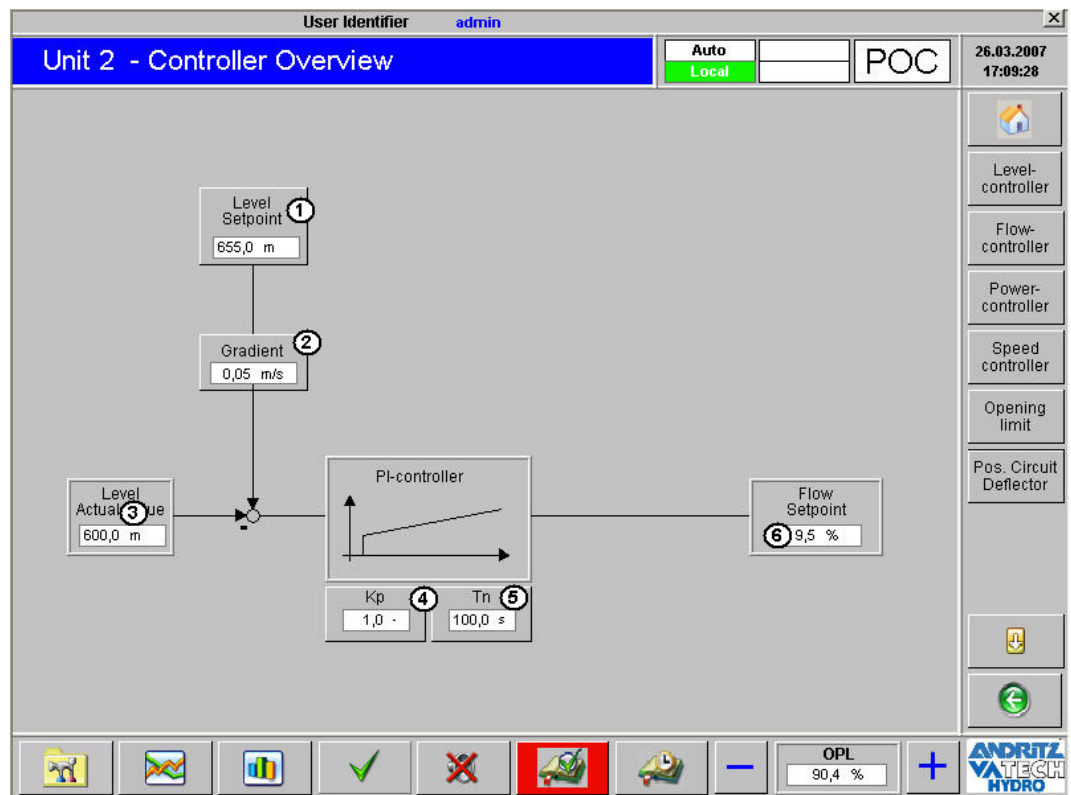


Fig. 6: Display – Level Controller

This detailed controller display constitutes the block diagram of the level controller. Besides the level setpoint (1), important parameters of this controller can also be modified here. They include the gradient for the readjustment of the level setpoint (2) as well as the controller parameters K_P and T_N (4 and 5).

Variables shown:

No.	Variable	Description
1	P.NW.AO.LSP	level setpoint [m]
2	P.NW.AO.GRD_LEC	gradient - level setpoint [m/s]
3	P.NW.AO.HHW1_INT	headwater level [m]
4	P.NW.AO.KP_LEC	proportional gain - level controller [-]
5	P.NW.AO.TN_LEC	Integral-action time - level controller [s]
6	P.NW.AO.FSP_LEC	output - level controller [%]

Tab 3: Variables – Display Level Controller

5.5 Display – Flow Controller

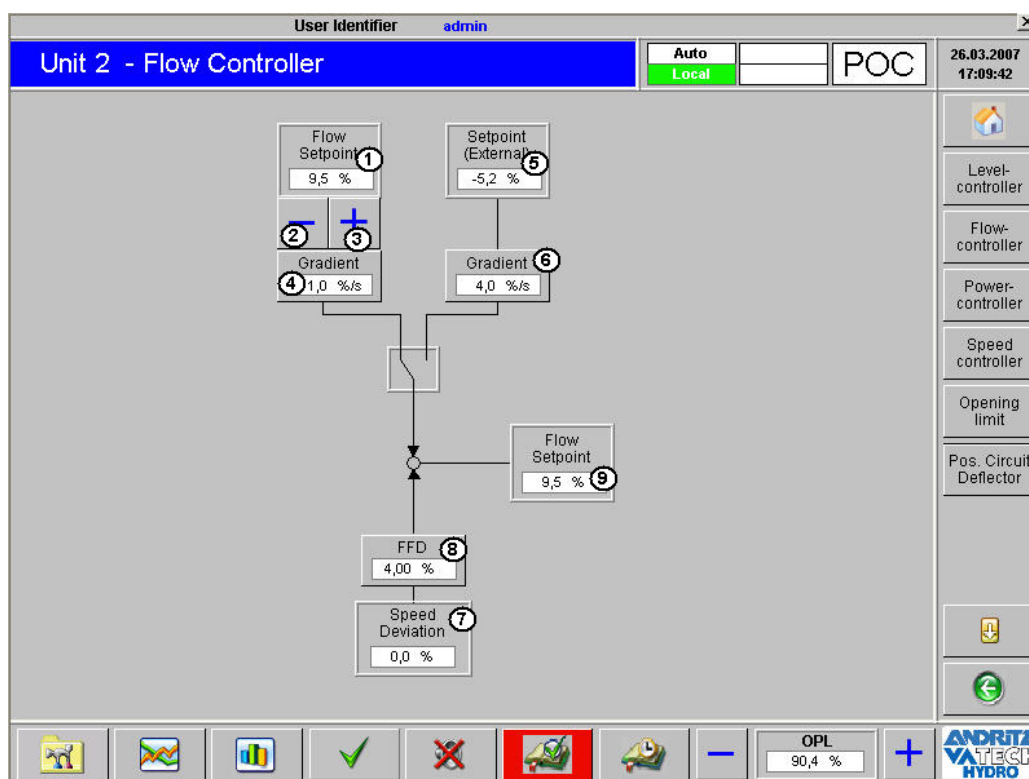


Fig. 7: Display – Flow Controller

This detailed controller display constitutes the block diagram of the flow controller. Besides the flow setpoint (1 to 3), important parameters of this controller can also be modified here. They include the gradient for the readjustment of the flow setpoint by using the (+) and (–) buttons (4), the gradient for the external flow setpoint value (6) as well as the frequency flow droop (8).

Variables shown:

No.	Variable	Description
1	P.NW.AO.FSP_%	internal flow setpoint [%]
2	P.NW.DI.PSP/FSP-	flow setpoint lower
3	P.NW.DI.PSP/FSP+	flow setpoint higher
4	P.NW.AO.TIM_FSP	gradient - flow setpoint (+/-) [%/s]
5	P.NW.AO.FSP_INT	external flow setpoint [%]
6	P.NW.AO.GRD_FSP	gradient - external flow setpoint [%/s]
7	P.NW.AO.E-F	speed deviation [%]
8	P.NW.AO.FFD	frequency flow droop [%]
9	P.NW.AO.FSP_FSP	output - flow controller [%]

Tab 4: Variables – Display Flow Controller

5.6 Display – Power Controller

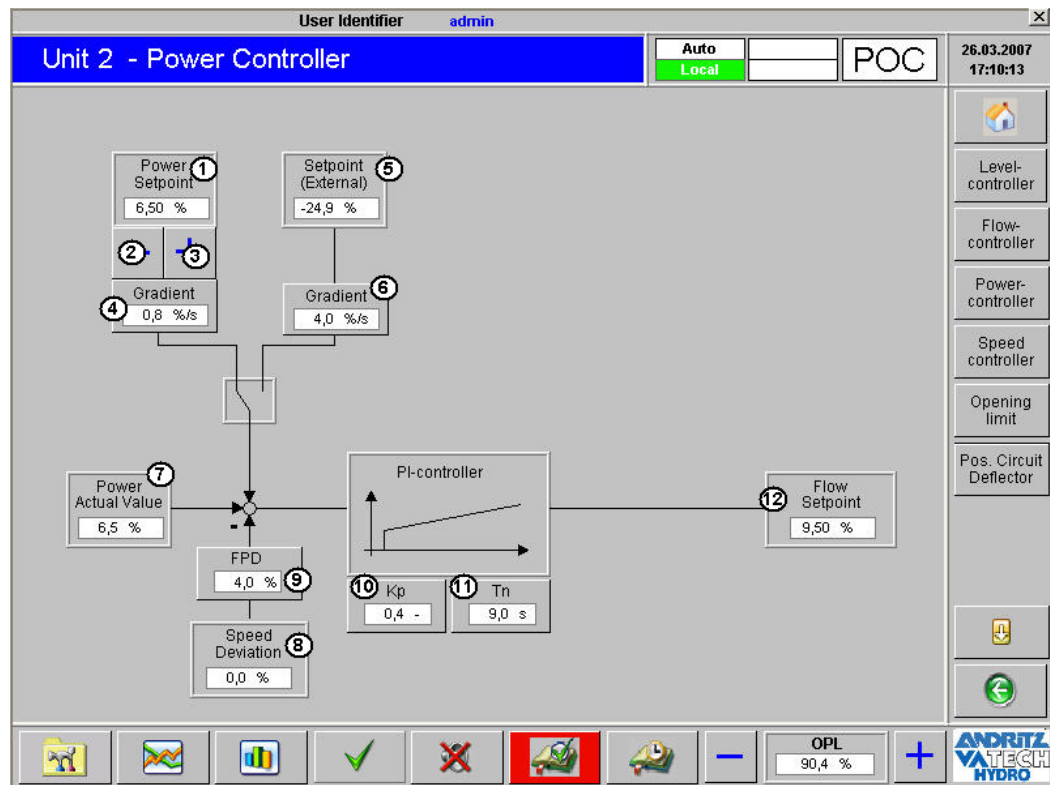


Fig. 8: Display – Power Controller

This detailed controller display constitutes the block diagram of the power controller. Besides the power setpoint (1 to 3), important parameters of this controller can also be modified here. They include the gradient for the readjustment of the power setpoint by using the (+) and (–) buttons (4), the gradient for the external power setpoint (6), the frequency power droop (9), as well as the controller parameters K_P and T_N (10 and 11).

Variables shown:

No.	Variable	Description
1	P.NW.AO.PSP_%	internal power setpoint [%]
2	P.NW.DI.PSP/FSP-	Leistungs-Sollwert tiefer
3	P.NW.DI.PSP/FSP+	Leistungs-Sollwert höher
4	P.NW.AO.TIM_PSP	power setpoint lower
5	P.NW.AO.PSP_INT	power setpoint higher
6	P.NW.AO.GRD_PSP	gradient - power setpoint (+/-) [%/s]
7	P.NW.AO.P_INT_%	active power [%][x1]
8	P.NW.AO.E-F	speed deviation [%]
9	P.NW.AO.FPD	frequency power droop [%]
10	P.NW.AO.KP_POC	proportional gain - power controller [-]
11	P.NW.AO.TN_POC	Integral-action time - power controller [s]
12	P.NW.AO.FSP_POC	output - power controller [%]

Tab 5: Variables – Display Power Controller

5.7 Display – Speed Controller

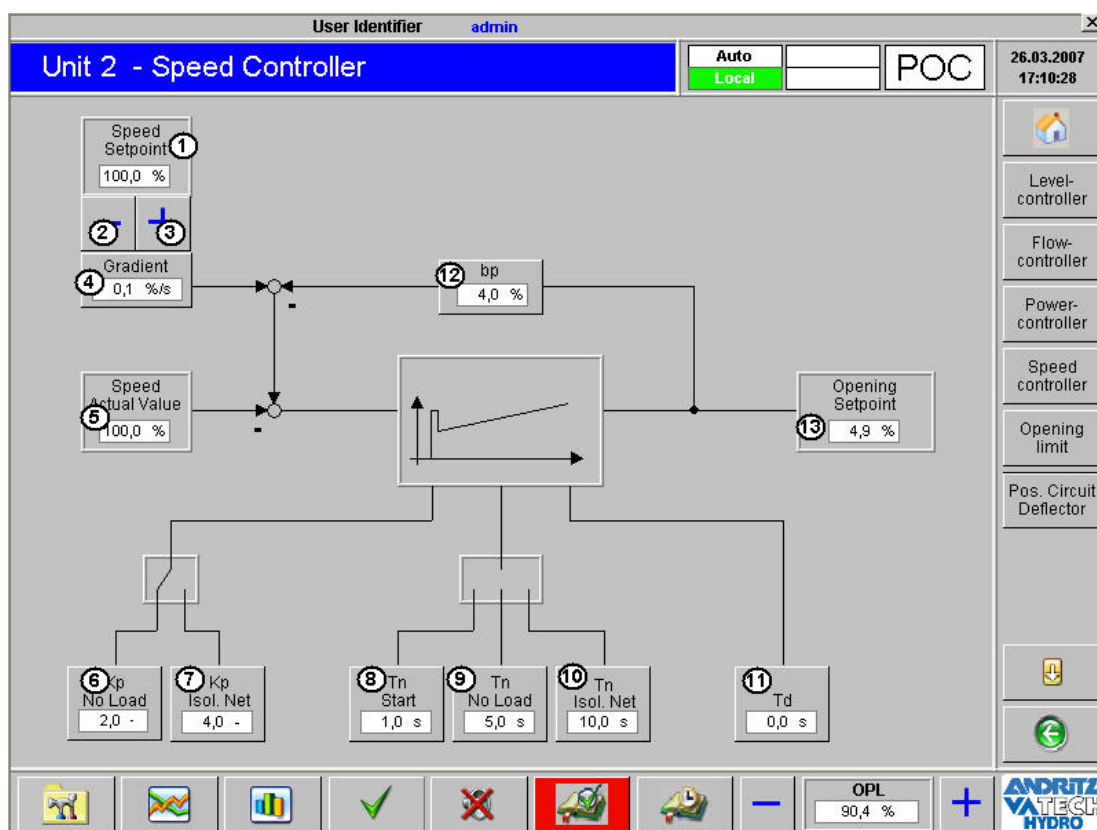


Fig. 9: Display – Speed Controller

This detailed controller display constitutes the block diagram of the speed controller. Contrary to the other controllers, various setpoint values can be defined and modified here as a function of current operating state (1 to 3, 5 and 6). The same applies to the controller parameters K_P and T_N (9 to 13). The controller parameter T_D (14) does not depend on the respective operating state. Furthermore, the gradient for the readjustment of the speed setpoint by using the (+) and (-) buttons (4) as well as the permanent speed droop (15) can be altered here.

Variables shown:

No.	Variable	Description
1	P.NW.AO.SSP	speed setpoint [%]
2	P.NW.DI.SSP-	speed setpoint lower
3	P.NW.DI.SSP+	speed setpoint higher
4	P.NW.AO.SSP_TIM	gradient - speed setpoint (+/-) [%/s]
5	P.NW.AO.n_ACTIVE	speed [%]
6	P.NW.AO.KP_NOLOAD	proportional gain - speed controller noload [-]
7	P.NW.AO.KP_ISNET	proportional gain - speed controller isolated network [-]
8	P.NW.AO.TN_START	integral-action time - speed controller start [s]
9	P.NW.AO.TN_NOLOAD	integral-action time - speed controller noload [s]
10	P.NW.AO.TN_ISNET	integral-action time - speed controller isolated network [s]
11	P.NW.AO.TD_SPC	Derivative-action time - speed controller [s]
12	P.NW.AO.bp_SPC	permanent droop - speed controller [%]
13	P.NW.AO.SPC_N-SP	output - speed controller [%]

Tab 6: Variables – Display Speed Controller

5.8 Display – Opening Limit

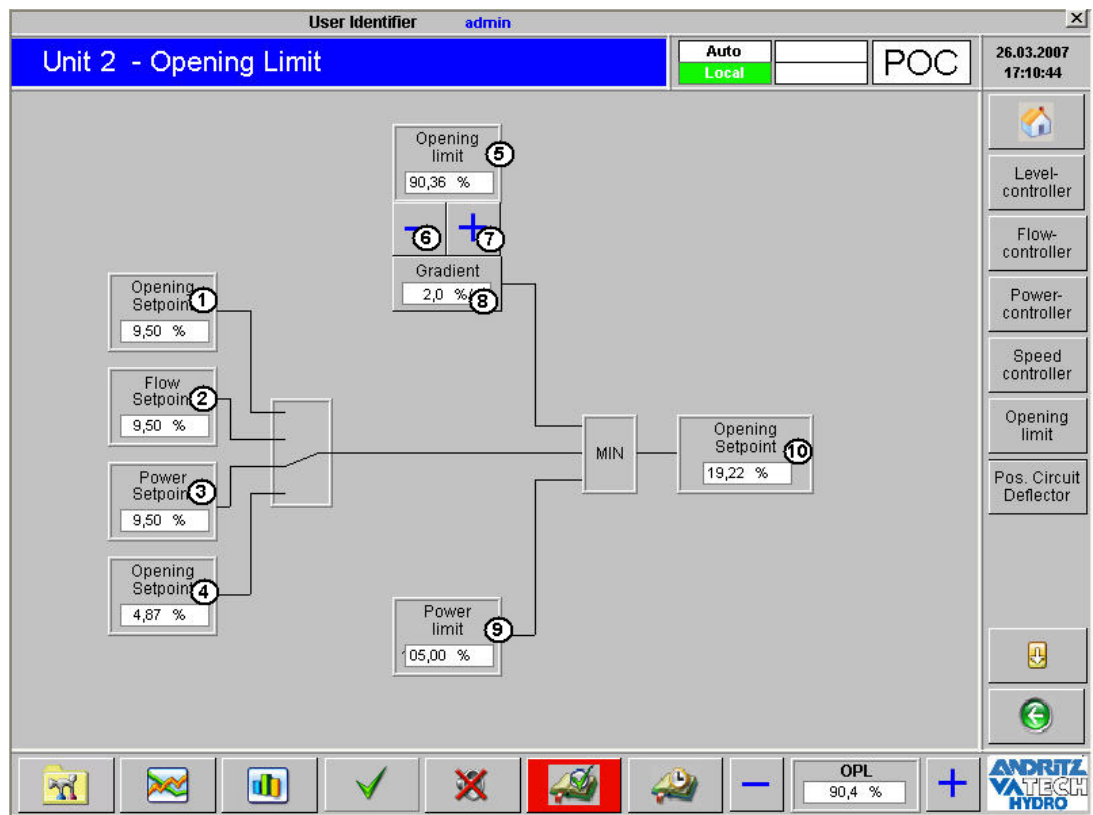


Fig. 10: Display – Opening Limit

This detailed display constitutes the block diagram of the opening limit. Besides the actual limitation (6 and 7) the associated gradient for the readjustment of this limit value (8) can be altered here. In addition, the maximum opening limitation (9) and the gradient for the increasing opening limitation (10) can be adjusted. This gradient be effective only in the case of abrupt changes of the opening limitation. The opening setpoint (12) corresponds to the setpoint of the positioning circuits described hereafter.

Variables shown:

No.	Variable	Description
1	P.NW.AO.FSP_LEC	output - level controller [%]
2	P.NW.AO.FSP_FSP	output - flow controller [%]
3	P.NW.AO.FSP_POC	output - power controller [%]
4	P.NW.AO.SPC_N-SP	output - speed controller [%]
5	P.NW.AO.OPL	opening limitation [%]
6	P.NW.DI.OPL-	opening setpoint lower
7	P.NW.DI.OPL+	opening setpoint higher
8	P.NW.AO.TIM_OPL	gradient - opening limitation (+/-) [%/s]
9	P.NW.AO.Y_LI_GOV	power limitation - generator [%]
10	P.NW.AO.SP_OPL_MAX	opening setpoint [%]

Tab 7: Variables – Display Opening Limiter

5.9 Display – Positioning Circuit Deflector

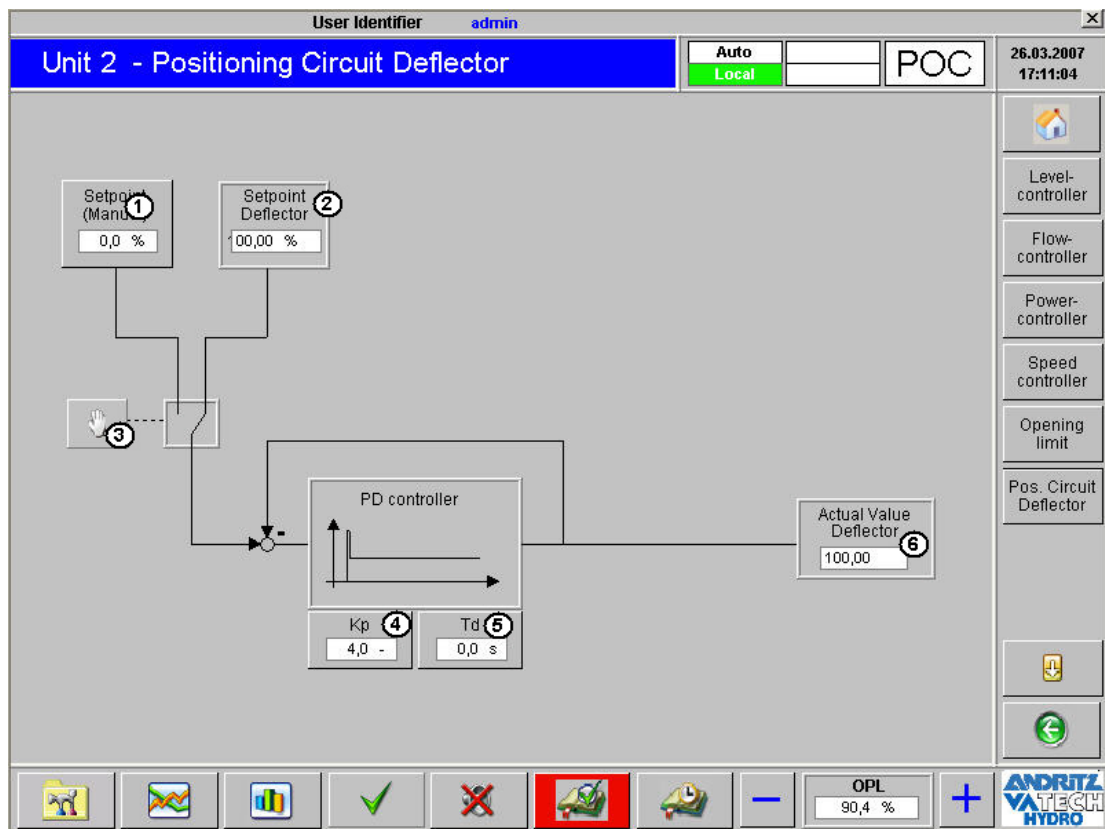


Fig. 11: Display – Positioning Circuit Deflector

This display represents the block diagram of the positioning circuit for the deflector. In this display, not only the controller parameters K_P and T_D (4 and 5) can be set, but also switchover to a manual setpoint for the positioning can be carried out (3). This manual setpoint can be specified individually (1). In addition, the tracking of the deflector can be switched off (7).

Variables shown:

No.	Variable	Description
1	P.NW.AO.MAN_DE_SP	manual setpoint - deflector [%]
2	P.NW.AO.DE-SP	setpoint - deflector [%]
3	P.NW.DI.MAN_DE	manual setpoint on
4	P.NW.AO.KP_DE	proportional gain - positioning controller deflector [-]
5	P.NW.AO.TD_DE	integral-action time - positioning controller deflector [s]
6	P.NW.AO.DE_INT	position - deflector [%]

Tab 8: Variables – Display Positioning Circuit Deflector

5.10 Display – Positioning Circuit Needle

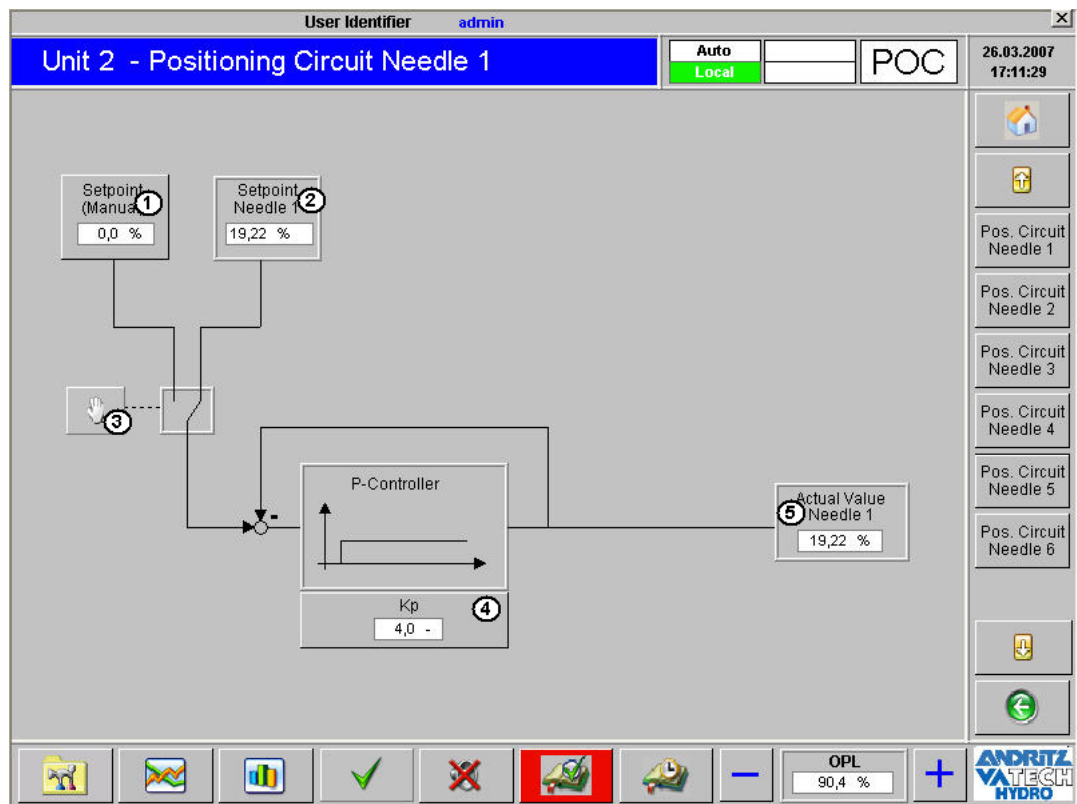


Fig. 12: Display – Positioning Circuit Needle

This display represents the block diagram of the positioning circuit for the needles. This display exists a total of six times in the visualization, according to the number of needles. All displays have the same construction, for example purposes the positioning circuit for needle 1 is presented here. In this display as well, not only the controller parameter K_p (4) can be set, but also switchover to a manual setpoint for the positioning can be carried out (3). This manual setpoint can be specified individually here too (1).

Variables shown:

No.	Variable	Description
1	P.NW.AO.MAN_N1_SP	manual setpoint - needle 1 [%]
2	P.NW.AO.N1-SP	setpoint - needle 1 [%]
3	P.NW.DI.MAN_N1	manual setpoint on
4	P.NW.AO.KP_N1	proportional gain - positioning controller needle 1 [-]
6	P.NW.AO.N1_INT	position - needle 1 [%]

Tab 9: Variables – Display Positioning Circuit Needle 1

5.11 Display – Needle Selection

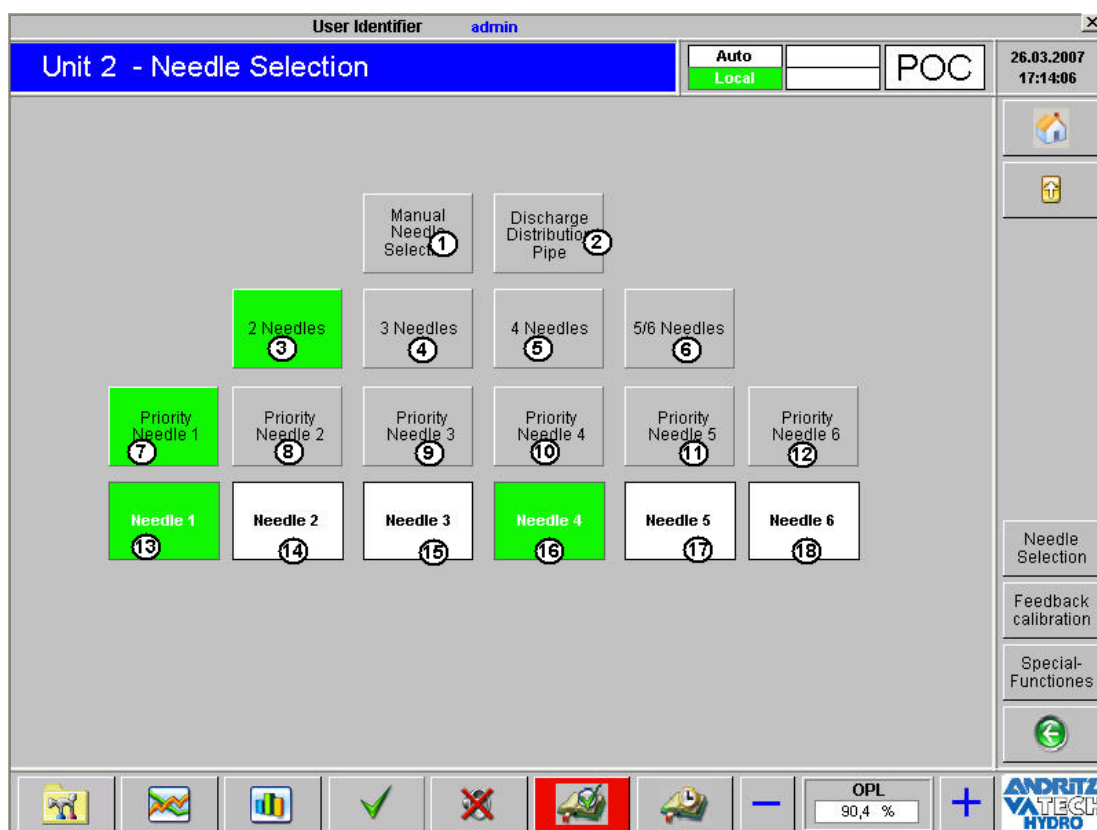


Fig. 13: Display – Needle Selection

This display is used for manual selection of the number of needles. After switching on the manual needle selection (1) the number of needles can be selected by choosing the corresponding option (2 to 5). The status of each individual needle is shown (6 to 11). In addition, it is also possible to discharge the ring feeder (12) on this display.

Variables shown:

No.	Variable	Description
1	P.NW.DI.MAN_NEEDLE P.NW.DO.MAN_NDL_ACT	Manual needle selection on Indication Manual needle selection on
2	P.NW.DI.N_DISCHARG P.NW.DO.N_DISCH_ACT	Discharge spiral case Indication Discharge spiral case
3	P.NW.DI.2N_ON/OFF P.NW.DO.2N_ACTIVE	Operation with 2 needles Indication Operation with 2 needles
3	P.NW.DI.3N_ON/OFF P.NW.DO.3N_ACTIVE	Operation with 3 needles Indication Operation with 3 needles
4	P.NW.DI.4N_ON/OFF P.NW.DO.4N_ACTIVE	Operation with 4 needles Indication Operation with 4 needles
5	P.NW.DI.5/6N_ON/OFF P.NW.DO.5/6N_ACTIVE	Operation with 5/6 needles Indication Operation with 5/6 needles
7-12	P.NW.DI.PRI_x P.NW.DO.PRI_x_ACT	Priority Needle x Indication Priority Needle x
13-18	P.NW.DO.Nx_ON_INT	Needle x active

Tab 10: Variables – Display of manual needle selection

5.12 Display – Feedback Calibration

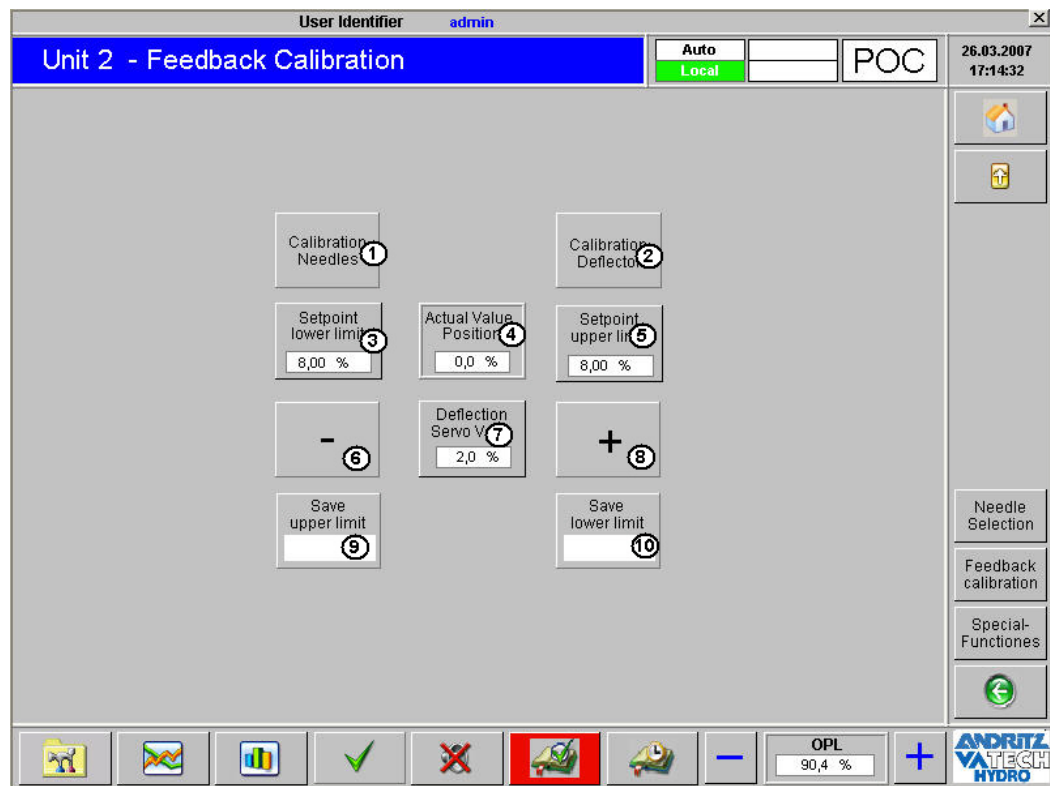


Fig. 14: Display – Feedback Calibration

Just like the commissioning display, this display is also intended for use by commissioning staff only. This is where one can calibrate the feedback sensors of the deflector servomotor and the needles servomotors. To do so, first select the feedback sensor to be calibrated by activating the appropriate switch (1 and 2). Then, set the desired lower and upper limits (3 and 5). With the help of the deflection (7) the opening or closing velocity can be influenced. Now, use the (+) and (-) buttons (6 and 8) to move the servomotor all the way to the desired end position. As soon as the servomotor has reached the desired end position, actuate the appropriate "save" button (9 or 10) to adjust the setpoint to the actual position (4). The successful saving process is confirmed by an indication (see 9 and 10).

The selector switches (1 to 7) are interlocked with one another, i.e. only one of them can be active at any one time. The same applies to the (+) and (-) switches (12 and 13) which are also reset automatically when the selector button is switched over. All switches are deactivated automatically when the user exits the display.

Variables shown:

No.	Variable	Description
1	P.NW.DI.ADJ_SM_Nx	feedback calibration - servomotor needle on
2	P.NW.DI.ADJ_SM_DE	feedback calibration - servomotor deflector on
3	P.NW.AI.FBC_VAL_L	reference value minimum [%]
4	P.NW.AO.POS	current position [%]
5	P.NW.AI.FBC_VAL_H	reference value maximum [%]
6	P.NW.DI.FBC_MINUS_SV	servomotor - close
7	P.NW.AO.DEFLECTION	opening/closing deflection - servovalve [%]
8	P.NW.DI.FBC_PLUS_SV	Servomotor - open
9	P.NW.DI.FBC_MIN_KEY P.NW.DO.LOW_OK	Calibration minimum position Calibration minimum position achieved
10	P.NW.DI.FBC_MAX_KEY P.NW.DO.HIGH_OK	Calibration maximum position Calibration maximum position achieved

Tab 11: Variables – Display Feedback Calibration

5.13 Display – Special functions

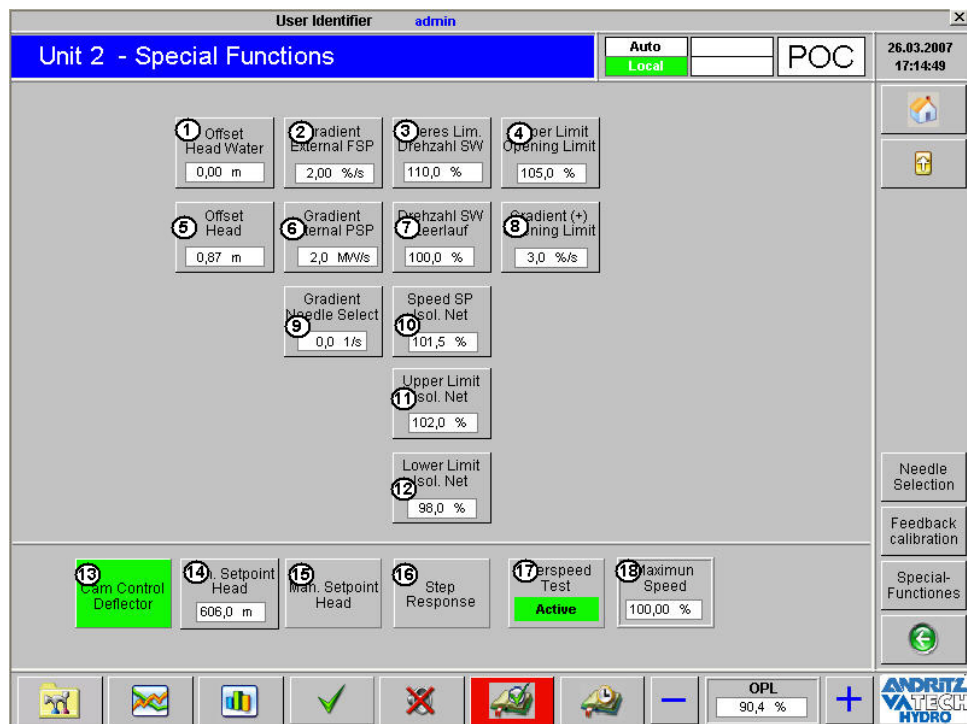


Fig. 15: Display – Commissioning

Just like the preceding one, this display is also intended for use by commissioning staff only. It can be used to adapt not only the offsets for the headwater level (1) and the head (2) and the gradients for the external setpoint values (6 and 7) but also the limits for isolated network detection (8 and 9) and the upper limit for the speed setpoint (11). The same applies for the gradient for smooth needle shifting (5). In addition a manual value for the head can be defined (3 and 4), an isolated network operation can be reset (10) and the overspeed test can be activated on (12).

Variables shown:

No.	Variable	Description
1	P.NW.AO.0-PT_HHW1	height correction value - headwater level [m]
2	P.NW.AO.GR_FSP	gradient for external flow setpoint value [%/s]
3	P.NW.AO.SSP_LIH	maximum value - speed setpoint [%]
4	P.NW.AO.LIH_OPL	Opening Limit high
5	P.NW.AO.0-PT_Hdyn	height correction value - pressure measurement head [m]
6	P.NW.AO.GR_PSP	gradient for external power setpoint value [%/s]
7	P.NW.AO.SSP_START	Speed setpoint no-load [%]
8	P.NW.AO.GRH_OPL	gradient + opening limit [%/s]
9	P.NW.AO.GRD_SHARE_0	gradient for smooth needle shifting [1/s]
10	P.NW.AO.SSP_ISNET	Speed setpoint island operation [%]
11	P.NW.AO.LIH_ISNET	upper limit value - isolated network detection [%]
12	P.NW.AO.LIL_ISNET	lower limit value - isolated network detection [%]
13	P.NW.DI.OCC_ON/OFF P.NW.DO.OCC_ACTIVE	Off Cam Control On/Off

Tab 12: Variables – Display special Functions

5.14 Display – Trend Diagram

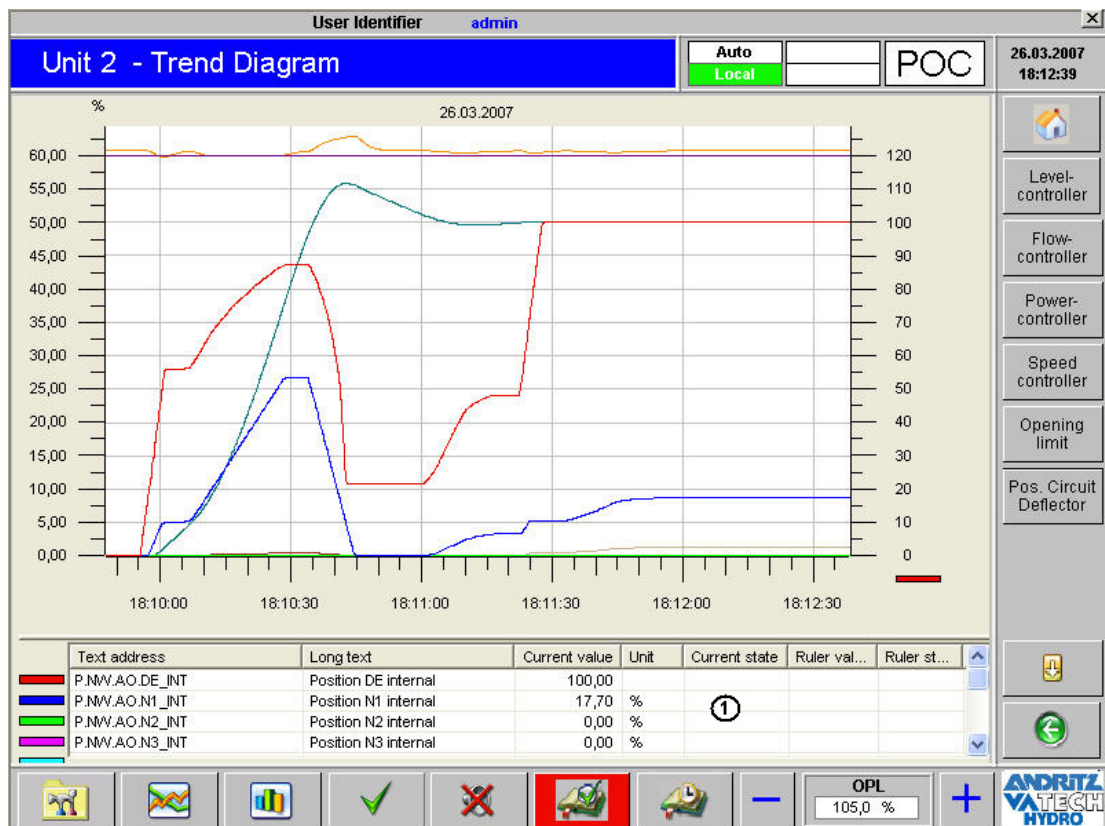


Fig. 16: Display – Trend Diagram

This diagram shows the chronological progress of the most important analog values of the machine. They are the same analog values that are also indicated on the main overview display and the measured value display. The legend pertaining to the curves including the measuring ranges and the curve colours is disposed above the diagram. In addition to the

known buttons, the supplementary menu contains a close button. Using this  button, the user can close the trend diagram, and the display from where the trend diagram was called up will be opened again.

Variables shown:

No.	Variable	Description
1	P.NW.AO.N_MAX	Maximum position needles [%]
2	P.NW.AO.DE_INT	position - deflector [%]
3	P.NW.AO.HHW1_INT	Oberwasserpegel [m]
4	P.NW.AO.H	head [m]
5	P.NW.AO.Q	flow [m³/s]
6	P.NW.AO.P_INT	active power [MW]
7	P.NW.AO.n_ACTIVE	speed [%]

Tab 13: Variables – Display Trend Diagram

5.15 Display – Measured Values

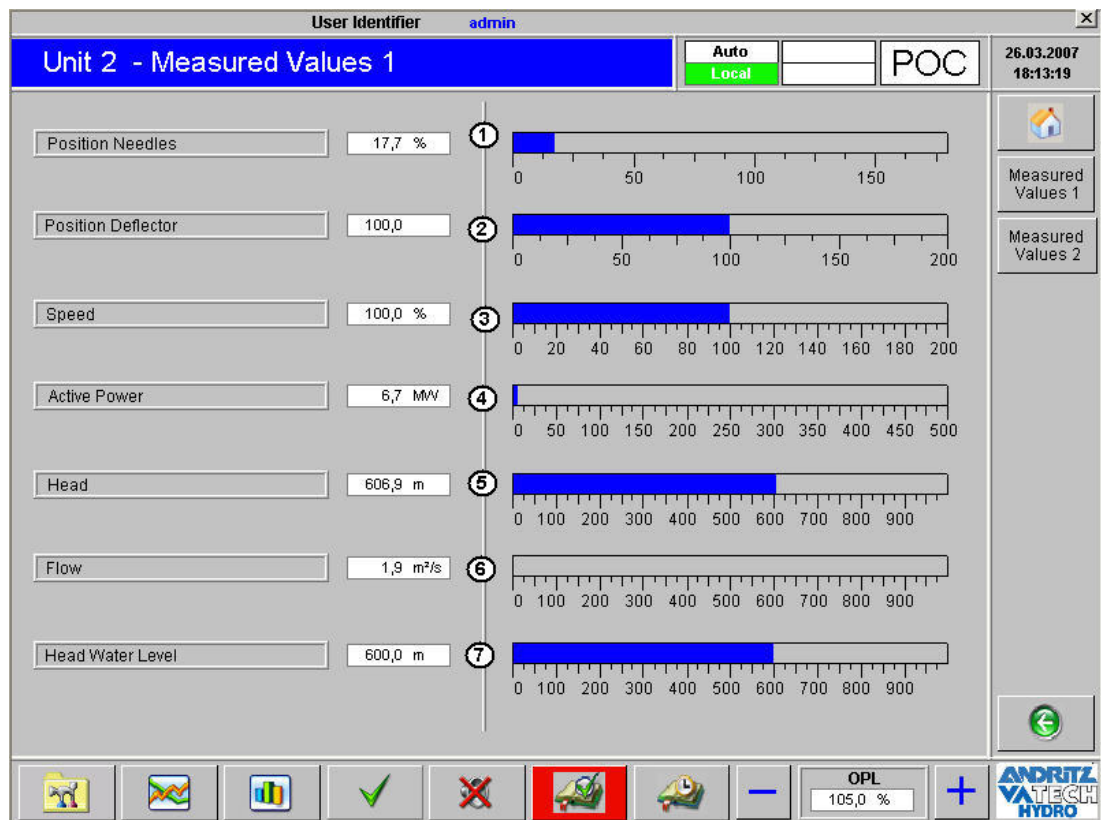


Fig. 17: Display – Measured Values

Also in this display, the most important analog values of the machine are shown. Their representation as bar diagrams is used for better clarity. What we are dealing with here are the same analog values as in the main overview display and the trend diagram.

In addition to the known buttons, the supplementary menu includes a close button. Using the  button, the user can close the trend diagram and open the display where the trend diagram had been called up from.

Variables shown:

No.	Variable	Description
1	P.NW.AO.N_MAX	Maximum position needles [%]
2	P.NW.AO.DE_INT	position - deflector [%]
3	P.NW.AO.n_ACTIVE	speed [%]
4	P.NW.AO.P_INT	active power [MW]
5	P.NW.AO.H	head [m]
6	P.NW.AO.Q	flow [m³/s]
7	P.NW.AO.HHW1_INT	headwater level [m]

Tab 14: Variables – Display Measured Values

5.16 Display - System Control

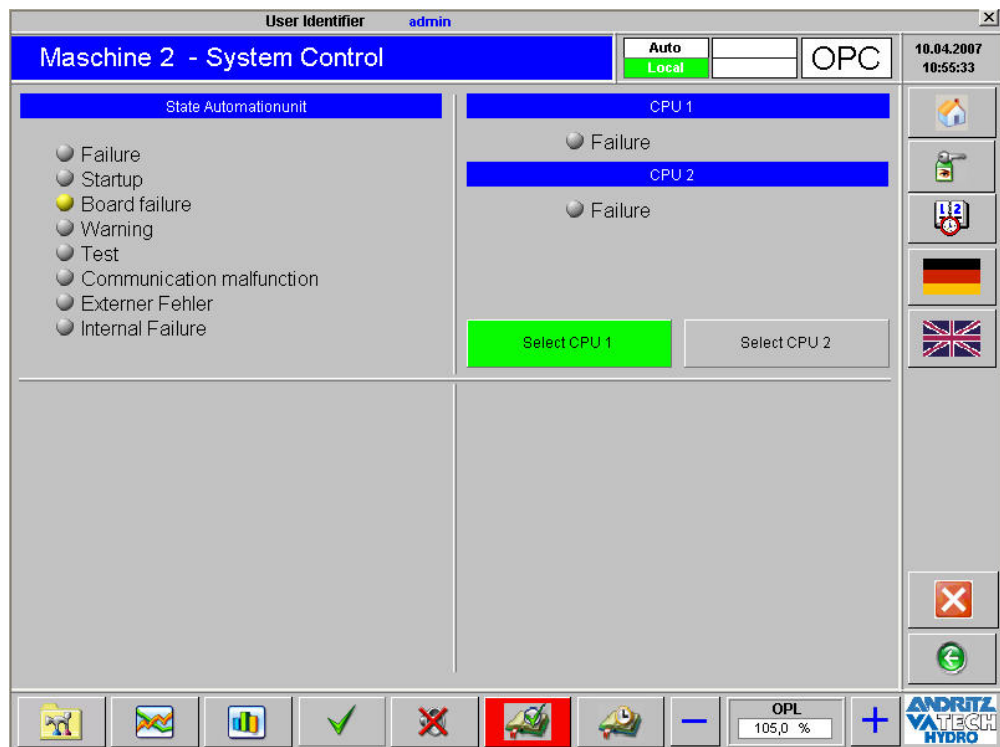


Fig. 18: Figure – System Control

This picture has an only main menu with additional functiones.

With the  button one can log-on to the system. This is carried out by means of the following dialog:

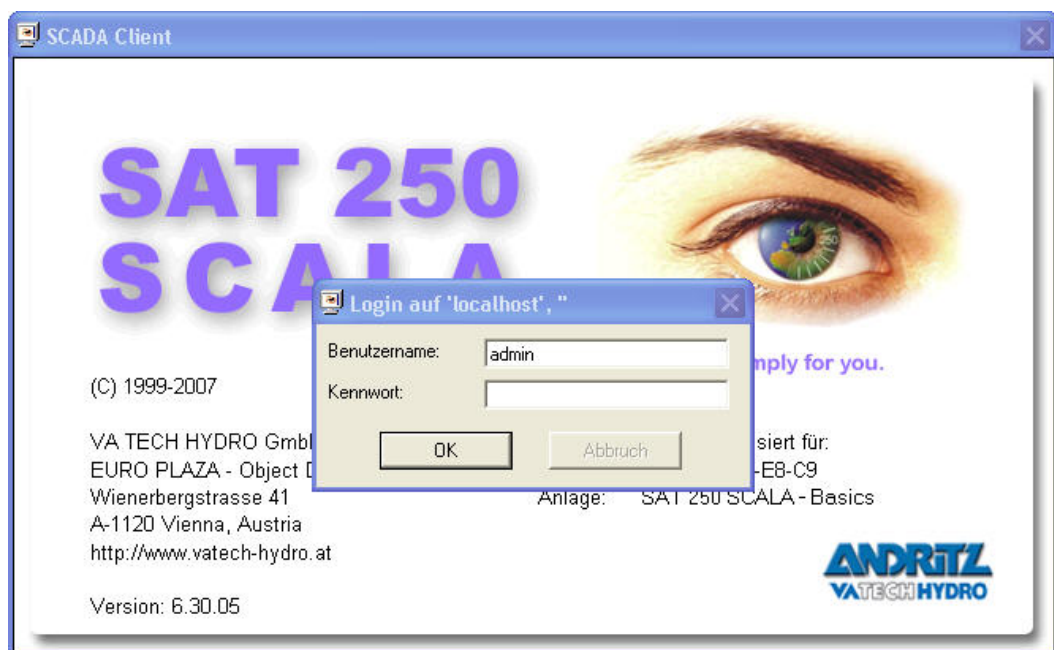


Abb. 19: Picture – Login

The input of the data is carried out using the virtual keypad on the Touch-Panel. After entering the user name and the associated password, correspondingly protected function calls or operator inputs can also be carried out. If the current authorizations for a particular function call or a particular operator input are insufficient, then with its selection the same login dialog is called up.

- Button operation



- log-on/off from the system



- terminate visualization (SAT250 runtime) program



- set date and time of touch panel



- close system control and open the window which was opened before

By pressing the  - button it's possible to set date and time for the touchpanel. The input of this setpoint is done via a virtual keyboard. By pressing the Cancel-button the input can be corrected. After pressing the O.K. -button date and time will be setted. By pressing Cancel all bevor entered values are discharged.

- Additionally also some important system states are indicate in this picture.

No.	Description
1	General Information about the automation unit
2	LAN state of the component
3	Internal use of memory
4	External use of memory (CF Card)
5	Current user
6	Software-version of SAT250 System
7	Software-version of SAT250 application
8	Software-version of CAEx plus application

Tab 15: Informations – systemcontrol

5.17 Display – Alarm Signal List

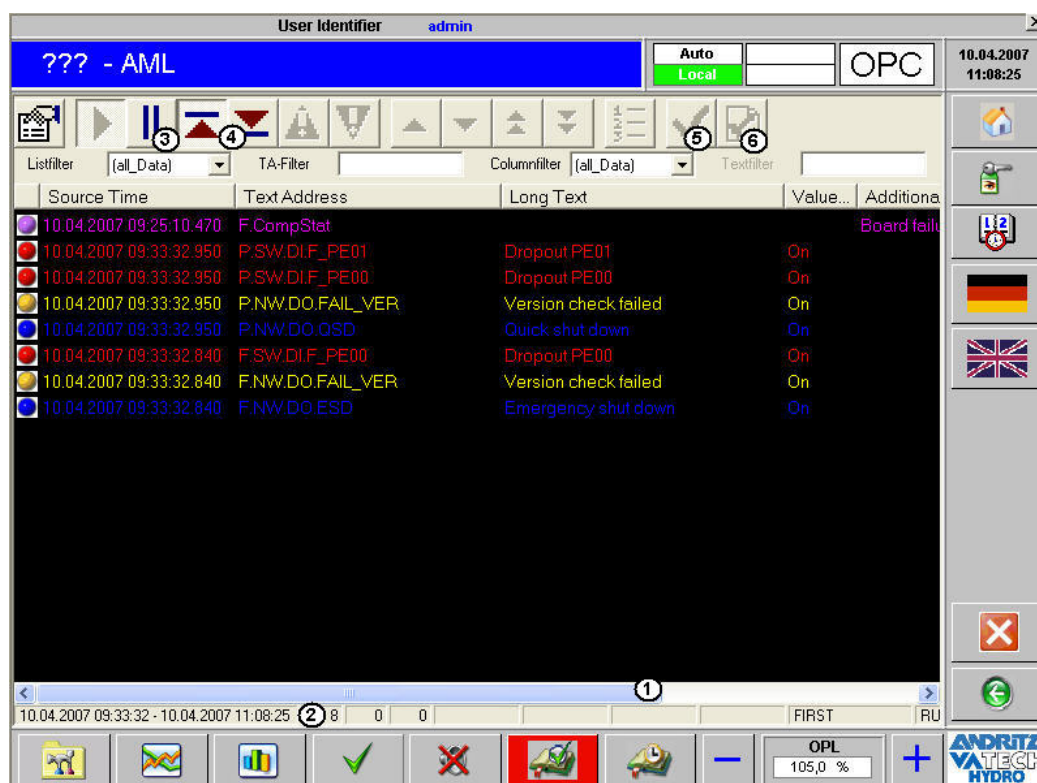


Fig. 20: Display – Alarm Signal List

With this display it has to do with the Alarm Signal List. In the list window all alarms (red), warnings (yellow) and trips (light blue) are listed in the corresponding color and with the associated identifier, the long text, the state, the time of occurrence, the time of acknowledgement and the time of going. The navigation within the list is carried out by means of the horizontal (1) and the, only visible if required, vertical slider. Under the list window there is a display of the number of alarms (2). In addition, one can acknowledge all alarms in the list (5) single quit and (6) for quitting all the alarms (8). An interruption of displaying the list can be done with (3) and (4)

Besides the familiar -button the supplementary menu contains a -button, with which the Alarm Signal List can be closed and that display opened from where one called up the Alarm Signal List.

5.18 Display – Chronological Event List

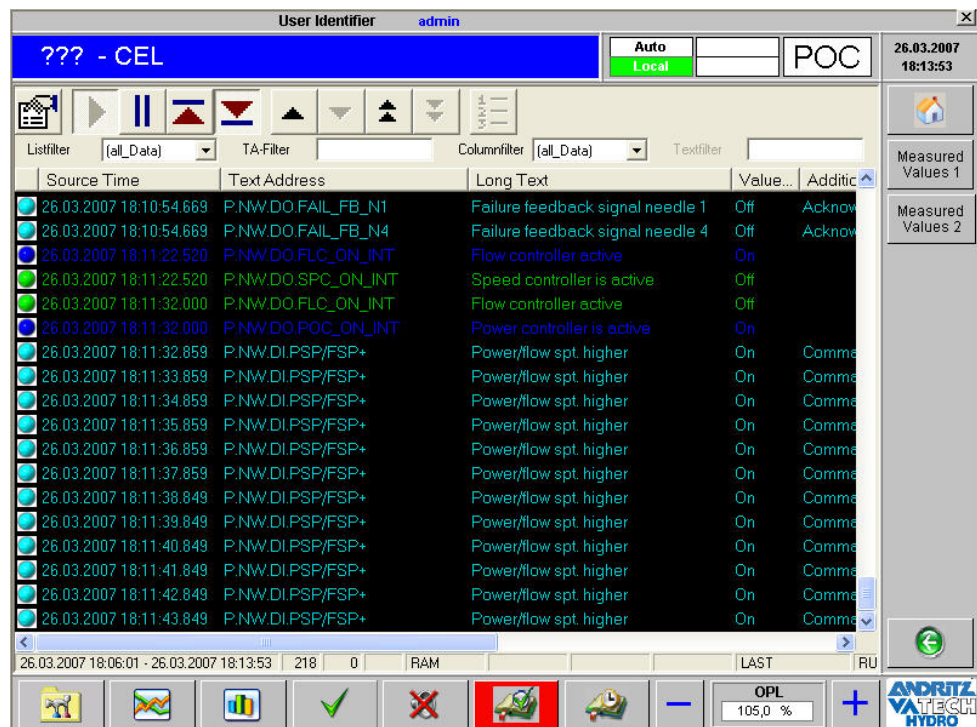


Fig. 21: Display – Chronologic Event List

With this display it has to do with the Chronologic Event-List. Here in the list window, besides all alarms (red), warnings (yellow) and trips (light blue), all binary information (green) as well as setpoint value changes and other operator inputs (dark blue) are also listed. As with the Alarm Signal List the listing is carried out in the corresponding color together with the associated identifier, the long text, the state and the time of the occurrence. The navigation within the list is carried out by means of the horizontal (1) and the, only visible if required, vertical slider (2). Below the list window there is a display of the number of list entries (3). In addition one can sort all entries in the list according to the time of their occurrence (4) as well as interrupt the recording of the events (5) and (6).

Besides the familiar -button the supplementary menu contains a -button, with which the Chronologic Event List can be closed and that display activated from where one called up the Chronological Event List.

6. Maintenance

6.1 Error List and Description of Cause

Error	Error name	Meaning of the error message	Possible cause	Error correction
P.TP.O.FAIL_DE	Failure periph. devices deflector	Fault – Peripheral devices Deflector → Quick shutdown	Open circuit, Defective sensor	Check cabling, Replace sensor
P.TP.O.FAIL_FSP	Failure ext. flow setpoint	Fault – External flow setpoint	Open circuit, Fault - Input module AI-6300	Check cabling, Replace module
P.TP.O.FAIL_Hdyn	Failure peripheral devices Hdyn	Fault – Peripheral device Pressure measuring	Open circuit, Defective sensor, Fault - Input module AI-6300	Check cabling, Replace sensor, Replace module
P.TP.O.FAIL_HHW1	Failure peripheral devices level	Fault – Peripheral device Level measuring	Open circuit, Defective sensor, Fault - Input module AI-6300	Check cabling, Replace sensor, Replace module
P.TP.O.FAIL_n_1	Failure speed Figk up 1	Fault – Speed sensor 1	Open circuit, Defective sensor, Fault - Input module TE-6420	Check cabling, Replace sensor, Replace module
P.TP.O.FAIL_n_2	Failure speed Figk up 2	Fault – Speed sensor 2	Open circuit, Defective sensor, Fault - Input module TE-6420	Check cabling, Replace sensor, Replace module
P.TP.O.FAIL_N1	Failure periph. devices needle 1	Fault – Peripheral devices Nozzle 1	Open circuit, Defective sensor	Check cabling, Replace sensor
P.TP.O.FAIL_N2	Failure periph. devices needle 2	Fault – Peripheral devices Nozzle 2	Open circuit, Defective sensor, Fault - Input module TE-6450, Fault - Output module AO-6380	Check cabling, Replace sensor, Replace module
P.TP.O.FAIL_N3	Failure periph. devices needle 3	Fault – Peripheral devices Nozzle 3	Open circuit, Defective sensor	Check cabling, Replace sensor
P.TP.O.FAIL_N4	Failure periph. devices needle 4	Fault – Peripheral devices Nozzle 4	Open circuit, Defective sensor, Fault - Input module TE-6450, Fault - Output module AO-6380	Check cabling, Replace sensor, Replace module
P.TP.O.FAIL_N5	Failure periph. devices needle 5	Fault – Peripheral devices Nozzle 5	Open circuit, Defective sensor	Check cabling, Replace sensor

Error	Error name	Meaning of the error message	Possible cause	Error correction
P.TP.O.FAIL_N6	Failure periph. devices needle 6	Fault – Peripheral devices Nozzle 6	Open circuit, Defective sensor, Fault - Input module TE-6450, Fault - Output module AO-6380	Check cabling, Replace sensor, Replace module
P.TP.O.FAIL_P	Failure active power	Fault – Active power signal	Open circuit, Fault - Input module AI-6300	Check cabling, Replace module
P.TP.O.FAIL_POS_DE	Failure deflector positioning circuit	Exceeding of the deviation of setpoint and actual positions – Deflector → Emergency shutdown	Fault - Input module TE-6450 or output module AO-6380, Open circuit, Fault - Servovalve, Fault - Feedback sensor	Check cabling, Replace module, Replace feedback sensor, Check hydraulics
P.TP.O.FAIL_POS_N1	Failure needle 1 positioning circuit	Exceeding of the deviation of setpoint and actual positions - Needle 1 → Quick shutdown	Fault - Input module TE-6450 or output module AO-6380, Open circuit, Fault - Servovalve, Fault - Feedback sensor, Jammed foreign bodies	Check cabling, Replace module, Replace feedback sensor, Check hydraulics, Remove jammed foreign bodies
P.TP.O.FAIL_POS_N2	Failure needle 2 positioning circuit	Exceeding of the deviation of setpoint and actual positions - Needle 2 → Quick shutdown	Fault - Input module TE-6450 or output module AO-6380, Open circuit, Fault - Servovalve, Fault - Feedback sensor, Jammed foreign bodies	Check cabling, Replace module, Replace feedback sensor, Check hydraulics, Remove jammed foreign bodies
P.TP.O.FAIL_POS_N3	Failure needle 3 positioning circuit	Exceeding of the deviation of setpoint and actual positions - Needle 3 → Quick shutdown	Fault - Input module TE-6450 or output module AO-6380, Open circuit, Fault - Servovalve, Fault - Feedback sensor, Jammed foreign bodies	Check cabling, Replace module, Replace feedback sensor, Check hydraulics, Remove jammed foreign bodies
P.TP.O.FAIL_POS_N4	Failure needle 4 positioning circuit	Exceeding of the deviation of setpoint and actual positions - Needle 4 → Quick shutdown	Fault - Input module TE-6450 or output module AO-6380, Open circuit, Fault - Servovalve, Fault - Feedback sensor, Jammed foreign bodies	Check cabling, Replace module, Replace feedback sensor, Check hydraulics, Remove jammed foreign bodies
P.TP.O.FAIL_POS_N5	Failure needle 5 positioning circuit	Exceeding of the deviation of setpoint and actual positions - Needle 5 → Quick shutdown	Fault - Input module TE-6450 or output module AO-6380, Open circuit, Fault - Servovalve, Fault - Feedback sensor, Jammed foreign bodies	Check cabling, Replace module, Replace feedback sensor, Check hydraulics, Remove jammed foreign bodies
P.TP.O.FAIL_POS_N6	Failure needle 6 positioning	Exceeding of the deviation of	Fault - Input module TE-6450	Check cabling, Replace

Error	Error name	Meaning of the error message	Possible cause	Error correction
	circuit	setpoint and actual positions - Needle 6 → Quick shutdown	or output module AO-6380, Open circuit, Fault - Servo valve, Fault – Feedback sensor, Jammed foreign bodies	module, Replace feedback sensor, Check hydraulics, Remove jammed foreign bodies
P.TP.O.FAIL_PSP	Failure ext. power setpoint	Fault – External power setpoint	Open circuit, Fault – Input module AI-6300	Check cabling, Replace module

6.2 Alarm/Signal List - "Trips"

Signal text	Description	Meaning	Action
1.3.MEY10.EH000.XM99	Dropout modul AO	Failure - Output moduleAO-6380	Quick shutdown
1.3.MEY10.EH001.XM00	Dropout deflector	Open circuit - Position sensor – Deflector	Quick shutdown
1.3.MEY10.EH001.XM99	Dropout modul SSI	Failure - Input module TE-6450	Quick shutdown
1.3.MEY10.EH002.XM99	Dropout modul SMS	Failure - Input module TE-6420	Quick shutdown
1.3.MEY10.EH004.XM99	Dropout modul DI	Failure - Output module DI-6100	Quick shutdown
1.3.MEY10.EH006.XM99	Dropout modul DO	Failure - Output module DO-6200	Quick shutdown
1.3.MEY10.EH009.XM99	Dropout PE00	Failure - Peripheral element PE00	Quick shutdown
P.TP.O.FAIL_DE	Failure periph. devices deflector	Fault - Peripheral devices - Deflector	Quick shutdown
P.TP.O.FAIL_POS_DE	Failure deflector positioning circuit	Exceeding of the deviation of setpoint and actual positions - Deflector	Emergency shutdown
P.TP.O.FAIL_POS_N1	Failure needle 1 positioning circuit	Exceeding of the deviation of setpoint and actual positions - Needle 1	Quick shutdown
P.TP.O.FAIL_POS_N2	Failure needle 2 positioning circuit	Exceeding of the deviation of setpoint and actual positions - Needle 2	Quick shutdown
P.TP.O.FAIL_POS_N3	Failure needle 3 positioning circuit	Exceeding of the deviation of setpoint and actual positions - Needle 3	Quick shutdown
P.TP.O.FAIL_POS_N4	Failure needle 4 positioning circuit	Exceeding of the deviation of setpoint and actual positions - Needle 4	Quick shutdown
P.TP.O.FAIL_POS_N5	Failure needle 5 positioning circuit	Exceeding of the deviation of setpoint and actual positions - Needle 5	Quick shutdown
P.TP.O.FAIL_POS_N6	Failure needle 6 positioning circuit	Exceeding of the deviation of setpoint and actual positions - Needle 6	Quick shutdown
P.TP.O.n>n3	Speed switch overspeed	Overspeed	Emergency shutdown

6.3 Alarm/Signal List - "Alarms"

Signal text	Description	Meaning	Action
P.TP.O.FAIL_n_1	Failure speed Figk up 1	Fault - Speed sensor 1	Alarm
P.TP.O.FAIL_n_2	Failure speed Figk up 2	Fault - Speed sensor 2	Alarm
P.TP.O.FAIL_P	Failure active power	Fault - Active power signal	Alarm

6.4 Alarm/Signal List - "Warnings"

Signal text	Description	Meaning	Action
1.3.MEY10.EH001.XM00	Dropout needle 1	Failure - Nozzle 1	Warning
1.3.MEY10.EH001.XM01	Dropout needle 3	Failure - Nozzle 3	Warning
1.3.MEY10.EH001.XM02	Dropout needle 5	Failure - Nozzle 5	Warning
1.3.MEY10.EH003.XM00	Dropout active power	Open circuit - Active power	Warning
1.3.MEY10.EH003.XM01	Dropout PSP	Open circuit - External power setpoint	Warning
1.3.MEY10.EH003.XM02	Dropout HHW1	Open circuit - Level measuring	Warning
1.3.MEY10.EH003.XM03	Dropout Hdyn	Open circuit - Pressure measuring	Warning
1.3.MEY10.EH003.XM99	Dropout modul AI	Failure - Input module AI-6300	Warning
1.3.MEY10.EH005.XM99	Dropout modul DI	Failure - Input module DI-6100	Warning
1.3.MEY10.EH010.XM99	Dropout modul AO	Failure - Output module AO-6380	Warning
1.3.MEY10.EH011.XM99	Dropout modul AO	Failure - Output module AO-6380	Warning
1.3.MEY10.EH012.XM00	Dropout needle 2	Failure - Nozzle 2	Warning
1.3.MEY10.EH012.XM01	Dropout needle 4	Failure - Nozzle 4	Warning
1.3.MEY10.EH012.XM02	Dropout needle 6	Failure - Nozzle 6	Warning
1.3.MEY10.EH012.XM99	Dropout modul SSI	Failure - Input module TE-6450	Warning
1.3.MEY10.EH013.XM00	Dropout FSP	Open circuit - External flow setpoint	Warning
1.3.MEY10.EH013.XM99	Dropout modul AI	Failure - Input module AI-6300	Warning
1.3.MEY10.EH014.XM99	Dropout modul DO	Failure - Output module DO-6200	Warning
1.3.MEY10.EH019.XM99	Dropout PE01	Failure - Peripheral element PE01	Warning
AK_R001_K003_00	Internal error	Internal error in the TC 1703	Warning
AK_R001_K003_01	External error	External error (e.g. module failure etc.)	Warning
AK_R001_K003_02	Communication error	Communications faulty	Warning
AK_R001_K003_04	Warning	Warning	Warning
AK_R001_K003_05	Unit failure	Module failure	Warning
AK_R001_K003_06	Component failure	TC 1703 failed	Warning
P.TP.O.FAIL_FSP	Failure ext. flow setpoint	Fault - External flow setpoint	Warning

Signal text	Description	Meaning	Action
P.TP.O.FAIL_Hdyn	Failure peripheral devices Hdyn	Fault - Peripheral device Pressure measuring	Warning
P.TP.O.FAIL_HHW1	Failure peripheral devices level 1	Fault - Peripheral device Level measuring	Warning
P.TP.O.FAIL_N1	Failure periph. devices needle 1	Fault - Peripheral devices Nozzle 1	Warning
P.TP.O.FAIL_N2	Failure periph. devices needle 2	Fault - Peripheral devices Nozzle 2	Warning
P.TP.O.FAIL_N3	Failure periph. devices needle 3	Fault - Peripheral devices Nozzle 3	Warning
P.TP.O.FAIL_N4	Failure periph. devices needle 4	Fault - Peripheral devices Nozzle 4	Warning
P.TP.O.FAIL_N5	Failure periph. devices needle 5	Fault - Peripheral devices Nozzle 5	Warning
P.TP.O.FAIL_N6	Failure periph. devices needle 6	Fault - Peripheral devices Nozzle 6	Warning
P.TP.O.FAIL_POS_N1	Failure needle 1 positioning circuit	Exceeding of the deviation of setpoint and actual positions - Needle 1	Warning
P.TP.O.FAIL_POS_N2	Failure needle 2 positioning circuit	Exceeding of the deviation of setpoint and actual positions - Needle 2	Warning
P.TP.O.FAIL_POS_N3	Failure needle 3 positioning circuit	Exceeding of the deviation of setpoint and actual positions - Needle 3	Warning
P.TP.O.FAIL_POS_N4	Failure needle 4 positioning circuit	Exceeding of the deviation of setpoint and actual positions - Needle 4	Warning
P.TP.O.FAIL_POS_N5	Failure needle 5 positioning circuit	Exceeding of the deviation of setpoint and actual positions - Needle 5	Warning
P.TP.O.FAIL_POS_N6	Failure needle 6 positioning circuit	Exceeding of the deviation of setpoint and actual positions - Needle 6	Warning
P.TP.O.FAIL_PSP	Failure ext. power setpoint	Fault - External power setpoint	Warning

6.5 Alarm/Signal List - "Messages"

Signal text	Description	Meaning	Action
AK_R001_K003_03	Test	Online test active	Message
AK_R001_K003_07	Start-up	TC 1703 start-up	Message
P.TP.O.ISNET	Isolated network (int. or ext.)	Isolated operation	Message
P.TP.O.TU_LIMITED	Turbine limited	Turbine limited	Message